

PMIC for LCD panels

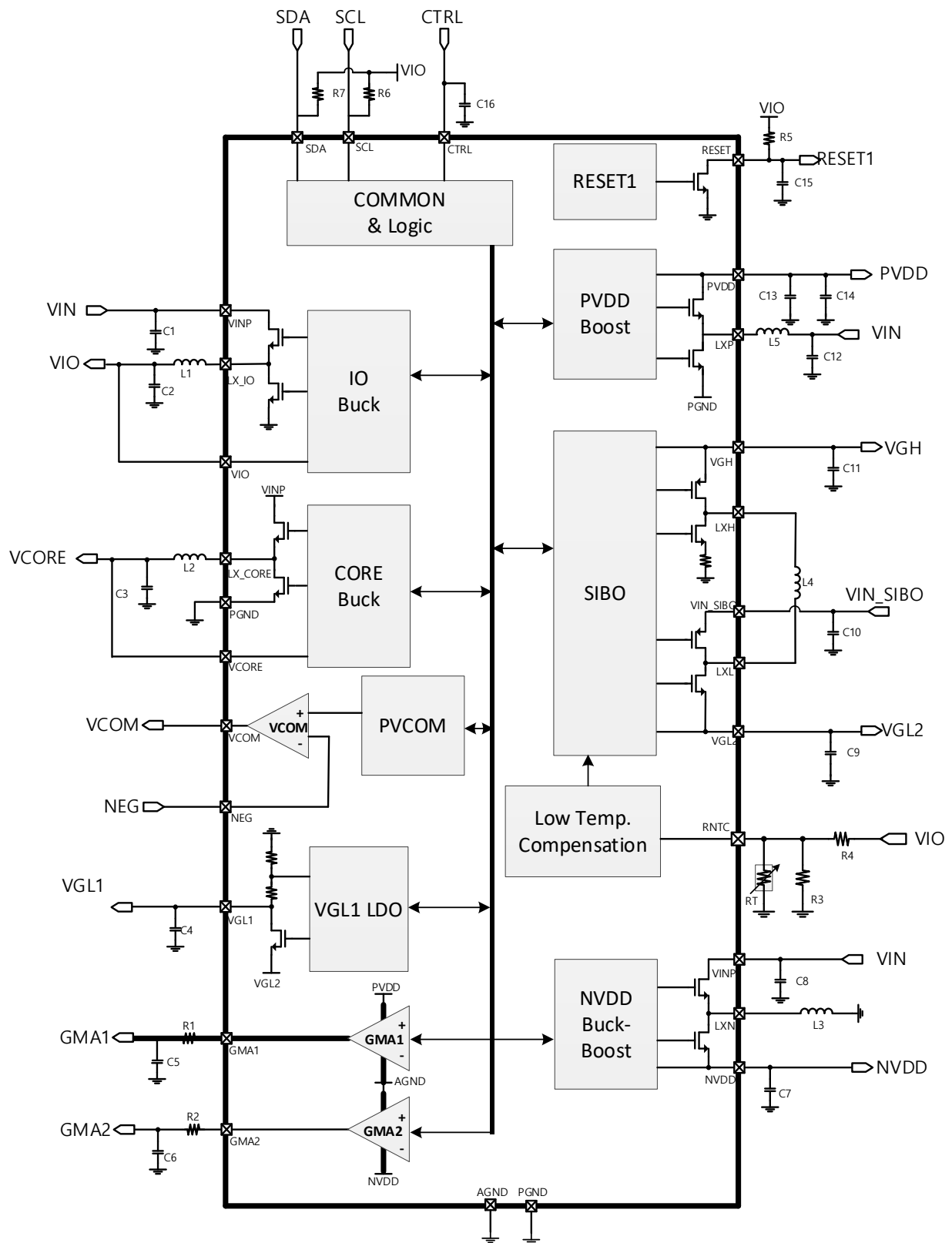
General Description

The LX52042C comprises a synchronous boost converter (PVDD), an inverting buck-boost converter (NVDD), a single-inductor bipolar-output (SIBO) converter for VGH and VGL2, a discharge, a programmable VCOM with an op-amp, two gamma buffers, a 2-ch buck converter, and one negative LDO(VGL1). The device in a FC-QFN 3.5X3.5-28L package is optimized for TFT LCD panels. The PVDD and NVDD converters provide the regulated positive and negative supply voltages for the source driver ICs of the TFT LCD panel, respectively. The VGH/VGL2 SIBO converter provides the supply voltages for the gate driver IC and level shifter. The PVDD, NVDD converters are the current-mode regulators. The high switching frequency allows the use of ultra-small inductors and ceramic capacitors. The current-mode control provides fast transient response to pulsed loads. The converters possess under-voltage lockout (UVLO), programmable soft-start, thermal protection, short circuit protection, and over current / voltage protection. The 2-ch buck converter provides logic voltages. The high-performance VCOM amplifier is designed as a VCOM buffer for the TFT LCD panels. The amplifier features a high short circuit current ($\pm 200\text{mA}$), wide bandwidth, and low voltage detector. The device also provides 8-bit adjustable reference voltages for gamma correction.

Features

- PVDD Boost Converter
 - 600kHz to 1.225MHz Switching Frequency
 - Built-In Low-Side MOSFET and High-Side MOSFET
 - Over Current Protection, Over Voltage Protection, Under Voltage Protection
 - Programmable Soft-Start
- NVDD Inverting Buck-Boost Converter
 - 600kHz to 1.225MHz Switching Frequency (Synchronous PVDD frequency)
 - Built-In Low-Side MOSFET and High-Side MOSFET
 - Over Current Protection, Over Voltage Protection, Under Voltage Protection
 - Programmable Soft-Start
- SIBO Converter for VGH and VGL2
 - 500kHz to 1.43MHz Switching Frequency
 - Built-In Low-Side MOSFET and High-Side MOSFET
 - Over Current Protection, Over Voltage Protection, Under Voltage Protection
 - Programmable Soft-Start
- Buck Converter
 - V_{CORE} : 0.8V to 2.06V, Min. 700mA
 - V_{IO} : 1.0V to 2.55V, Min. 300mA
- Negative LDO
 - V_{GL1} : -16V to -5.4V
 - 6-bit (64-Step) Adjustable Output
- Programmable Positive/Negative VCOM Amplifier
 - V_{COM} : -3.6 to 2.02V
 - $\pm 200\text{mA}$ Short Circuit Current
- Programmable 2-CH Gamma
 - $\pm 20\text{mA}$ Short Circuit Current
 - 6-bit (64-Step) Adjustable Output
- Low Voltage Detector
- Over Temperature Protection(OTP)
- Packaging Information
 - 28-pin, 3.5mm x 3.5mm, FC-QFN Package

Functional Block Diagram



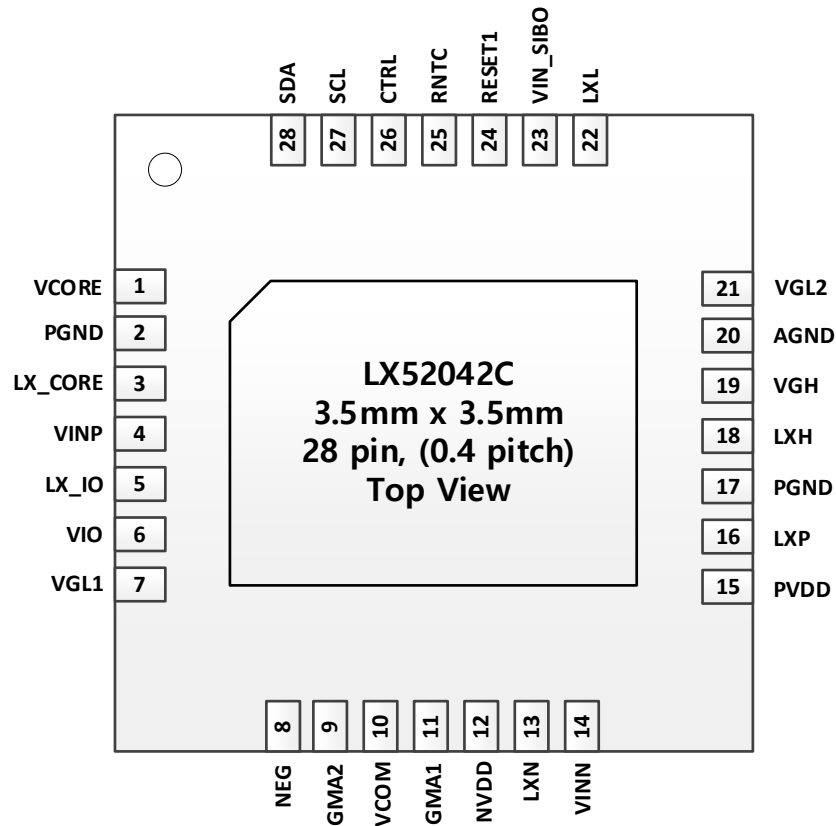
Recommended Component Values

INDEX	Description	VALUE	INDEX	Description	VALUE	Property
L1	VIO Inductor	4.7μH	C1	IO Buck VIN Cap	10μF	
L2	VCORE Inductor	4.7μH	C2	VIO Cap	10μF X 2ea	
L3	NVDD Inductor	4.7μH	C3	VCORE Cap	10μF	
L4	SIBO Inductor	4.7μH	C4	VGL1 Cap	2.2μF	
L5	PVDD Inductor	4.7μH	C5	GMA1 Cap	1μF	
R1	GMA1 Res.	0R	C6	GMA2 Cap	1μF	
R2	GMA2 Res.	0R	C7	NVDD Cap	10μF X 2ea	
R3	RNTC Res..	User setting value	C8	NVDD VIN Cap	10μF	
R4	RNTC Res.	User setting value	C9	VGL2 Cap	2.2μF	
R5	RESET1 Res.	4.7kΩ	C10	SIBO VIN Cap	10μF	
R6	SCL Res..	4.7kΩ	C11	VGH Cap	2.2μF	
R7	SDA Res.	4.7kΩ	C12	PVDD VIN Cap	10μF X 2ea	
			C13 C14	PVDD Cap	10μF X 3ea	
			C15	RESET1 Cap	NC	
			C16	CTRL Cap	0.1μF	

Note 1) These values should be optimized according to the operating condition of the user.

Pin Configuration and Description

(Top View)



NO	NAME	I/O	DESCRIPTION
E-PAD	PGND	POWER	Ground
1	VCORE	IN	VCORE buck converter output sensing pin
2	PGND	POWER	Ground
3	LX_CORE	IN/OUT	VCORE buck converter switching pin
4	VINP	POWER	Supply input power
5	LX_IO	IN/OUT	VIO buck converter switching pin
6	VIO	POWER	VIO buck converter output sensing pin
7	VGL1	IN/OUT	Negative LDO output pin
8	NEG	IN	VCOM amp negative input
9	GMA2	OUT	Gamma negative output
10	VCOM	OUT	VCOM amp output
11	GMA1	OUT	Gamma positive output
12	NVDD	POWER	NVDD buck-boost converter output
13	LXN	IN/OUT	NVDD buck-boost converter switching pin
14	VINN	POWER	Supply input power
15	PVDD	POWER	PVDD boost converter output and sensing pin
16	LXP	IN/OUT	PVDD boost converter switching pin
17	PGND	POWER	Ground
18	LXH	IN/OUT	VGH SIBO converter switching pin

19	VGH	IN/OUT	VGH SIBO converter output and sensing pin
20	AGND	POWER	Ground
21	VGL2	POWER	VGL2 SIBO converter output and sensing pin
22	LXL	IN/OUT	VGL2 SIBO converter switching pin
23	VIN_SIBO	POWER	SIBO supply input power
24	RESET1	OUT	RESET1 signal output pin
25	RNTC	IN	Low temperature compensation pin
26	CTRL	IN	Control Pin of Common logic
27	SCL	IN	Clock input for the I2C serial interface
28	SDA	IN/OUT	Data input/output for the I2C serial interface

Absolute Maximum Ratings**T_A = 25°C, unless otherwise noted**

PARAMETER		MIN	MAX	UNIT
PGND to AGND		-0.3	+0.3	V
VINP, VINN, SCL, SDA, RESET, RNTC, LX_IO, LX_CORE, VIO, VCORE to AGND		-0.3	+5.5	V
LX_CORE, LX_IO, to VIN		-5.5	+0.3	V
PVDD, LXP to PGND		-0.3	+7.7	V
LXP, GMA1 to PVDD		-7.7	+0.3	V
GMA1 to AGND		-0.3	+7.7	V
NVDD, GMA2 to AGND		-7.7	+0.3	V
LXN to AGND		-7.7	+5.5	V
LXN to VINN		-13.2	+0.3	V
LXN to NVDD		-0.3	+13.2	V
GMA2 to NVDD		-0.3	+7.7	V
VCOM, NEG to AGND		-7.7	+5.5	V
VCOM to VIN		-13.2	+0.3	V
VCOM to NVDD		-0.3	+13.2	V
VIN_SIBO to AGND		-0.3	+7.7	V
VGL2,VGL1 to AGND		-19.8	+0.3	V
VGL2,VGL1 to NVDD		-13.2	+0.3	V
LXL to VGL2		-0.3	+19.8	V
LXL to VIN_SIBO		-19.8	+0.3	V
VGH, LXH to AGND		-0.3	+33	V
LXH to VGH		-33	+0.3	V
Package Thermal Resistance(θ_{JA})		-	TBD	°C/W
Operating Ambient Temperature Range		-25	+85	°C
Operating Junction Temperature Range		-25	+125	°C
Storage Temperature Range		-65	+150	°C
Lead Temperature (Soldering, 10s)		-	+260	°C
ESD Rating	HBM (Human Body Model)	2000	-	V
	CDM (Charged Device Model)	700	-	V

Stresses beyond those listed under “Absolute Maximum Ratings” may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these or any other conditions beyond those indicated under “recommended operating conditions” is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

Electrical Characteristics

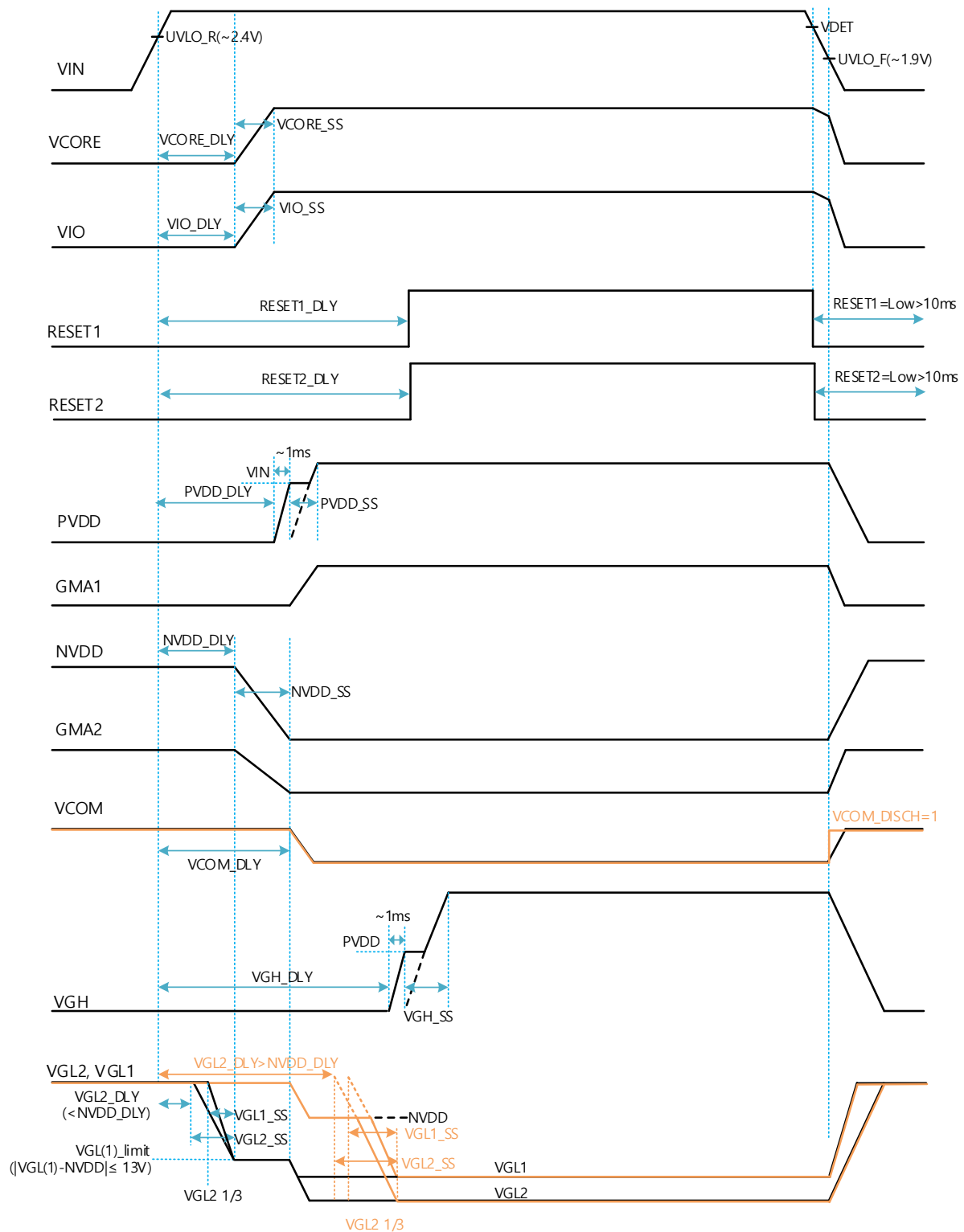
$V_{IN}=V_{INP}=V_{INN}=3.3V$, $V_{IN_SIBO}=PVDD=5V$, $NVDD=-5V$, $VGH=18V$, $VGL2=-12V$, $VGL1=-9V$, $V_{CORE}=1.2V$, $V_{IO}=1.8V$, $GMA1=4.8V$, $GMA2=-4.8V$, $PGND=AGND=0V$, $Freq=715kHz$, All Grounds are 0V, $T_A=25^{\circ}C$ (unless otherwise noted)

GENERAL						
PARAMETER	SYMBOL	CONDITION	MIN	TYP	MAX	UNIT
VIN Supply Voltage Range	V _{IN}	Normal operation, V _{IN} ≤ V _{PVDD} X 0.9	2.5	-	5	V
VIN Quiescent Current	I _{Q_VIN}	Converters not switching	-	0.5	1	mA
VIN Under-Voltage Lockout Threshold	V _{UVLO}	Vin rising	2.3	2.4	2.5	V
		Hysteresis	-	0.5	-	
OTP Threshold	T _{OTP_R}	Temperature rising	-	150	-	°C
	T _{OTP_HYS}	Hysteresis	-	20	-	
Current Mode Boost Converter - PVDD						
Operating Frequency	f _{PNVDD}	600/715/800/933/1000/1225kHz	600	-	1225	kHz
Frequency Accuracy		f _{PNVDD} =715kHz	-5		+5	%
		Others	-10		+10	%
Maximum Duty Cycle	D _{MAX_PVDD}			85	-	%
Under Voltage Protection	V _{UVP_PVDD}	V _{PVDD} falling under Target percentage	-	80	-	%
PVDD Fault Condition		UVP continued occur	-	50	-	ms
Output Voltage Range	V _{PVDD}	I2C control, V _{IN} ≤ V _{PVDD} 50mV/step	4.0	-	7.0	V
Output Voltage Accuracy	V _{ACC_PVDD}	no load, V _{PVDD} =5V	-1		+1	%
Load Regulation		V _{IN} =3.3V, V _{PVDD} =5V, I _{LOAD} =1 to 500mA, f _{PNVDD} =715kHz	-1		+1	%
Line Regulation		V _{IN} =2.5V~4.5V, I _{LOAD} =10mA	-1		+1	%
Current Limit	I _{LIM_PVDD}	1.5A/2.0A/2.5A/3.0A	1.5	-	3	A
Output Current Capability	I _{OUT_PVDD}	V _{IN} =3.3V, V _{PVDD} =5V	500	-	-	mA
Output Voltage – NVDD						
Operating Frequency	f _{PNVDD}	600/715/800/933/1000/1225kHz	-	F _{PNVDD}	-	Hz
Frequency Accuracy		f _{PNVDD} =715kHz	-5		+5	%
		Others	-10		+10	%
Output Voltage Range	V _{NVDD}	50mV/step	-7.15	-	-4.0	V
Output Voltage Accuracy	V _{ACC_NVDD}	no load, V _{NVDD} =5V	-1		+1	%
Under Voltage Protection	V _{UVP_NVDD}	V _{NVDD} falling under target	-	80	-	%
NVDD Fault Condition		UVP continued occur	-	50	-	ms
Load Regulation (Buck-Boost)	REG _{LOAD}	V _{IN} =3.3V, V _{NVDD} =-5V, I _{LOAD} =1 to 500mA, f _{PNVDD} =715kHz	-1		+1	%
Line Regulation	REG _{LINE}	V _{IN} =2.5V~5.0V, V _{NVDD} =-5V I _{LOAD} =10mA	-1		+1	%
Current Limit	I _{LIM_NVDD}	1.0A/1.5A/2.0A/2.5A	1.0	-	2.5	A
Output Current Capability	I _{OUT_NVDD}	V _{IN} =3.3V, V _{NVDD} =-5V	500	-	-	mA
Current Mode Buck Converter 1 - V _{CORE}						
Operating Frequency	f _{VCORE}	600/715/800/933/1000/1225kHz	600	-	1225	kHz
Frequency Accuracy		f _{VCORE} =715kHz	-5	-	+5	%
		Others	-10	-	+10	%

V _{CORE} Output Voltage	V _{CORE}	20mV/step	0.8	-	2.06	V
Output Voltage Accuracy	V _{ACC_VCORE}	no load, V _{CORE} =1.2V	-1	-	+1	%
Under Voltage Protection	V _{UVP_VCORE}	V _{CORE} falling under target	-	80	-	%
V _{CORE} Fault Condition		UVP continued occur	-	50	-	ms
Load Regulation	REG _{LOAD}	V _{CORE} =1.2V, I _{LOAD} =10 to 700mA	-1		+1	%
Line Regulation	REG _{LINE}	V _{IN} =2.5V to 5V	-1		+1	%
Current Limit	I _{LIM_CORE}		-	1.2	-	A
Maximum Driving Ability		V _{in} =3.3V, V _{CORE} =1.2V, f _{VCORE} =1225kHz	700	-	-	mA
Current Mode Buck Converter 2 – V_{IO}						
Operating Frequency	f _{VIO}	600/715/800/933/1000/1225kHz	600	-	1225	kHz
Frequency Accuracy		f _{VIO} =715kHz	-5	-	+5	%
		Others	-10	-	+10	%
V _{IO} Output Voltage	V _{IO}	50mV/step	1	-	2.55	V
Output Voltage Accuracy	V _{ACC_VIO}	no load, V _{IO} =1.8V	-1		+1	%
Under Voltage Protection	V _{UVP_VIO}	V _{IO} falling under target	-	80	-	%
V _{IO} Fault Condition		UVP continued occur	-	50	-	ms
Load Regulation	REG _{LOAD}	V _{IO} =1.8V, I _{LOAD} =10 to 300mA	-1	-	1	%
Line Regulation	REG _{LINE}	V _{IO} =1.8V, V _{IN} =2.5V to 5V	-1		+1	%
Current Limit	I _{LIM_IO}			1.2		A
Maximum Driving Ability		V _{in} =3.3V, V _{IO} =1.8V, f _{VIO} =1225kHz	300	-	-	mA
Negative LDO Regulator – V_{GL1}						
LDO Output Voltage	V _{VGL1}	200mV/step	-16	-	-5.4	V
Output Voltage Accuracy	V _{ACC_VGL1}	no load, V _{VGL1} =-12V	-3		+3	%
Dropout Voltage	V _{DO_VGL1}	I _{LOAD} =10mA		0.5		V
Output Current Capability	I _{OUT_VGL1}	V _{VGL1} =-12V	10	-	-	mA
Under Voltage Protection	V _{UVP_VGL1}		-	NVDD-1	-	V
V _{GL1} Fault Condition		UVP continued occur	-	50	-	ms
SIBO Converter - V_{GH}						
Operating Frequency	f _{SIBO}	500/600/715/800/933/1000 /1225/1430kHz	500	-	1430	kHz
Frequency Accuracy		f _{SIBO} =715kHz	-5		+5	%
		Others	-10		+10	%
Output Voltage Range	V _{VGH}	200mV/step, V _{GH_High} Resolution=0	6	-	12	V
		500mV/step, V _{GH_High} Resolution=1	12.5	-	28	
Output Voltage Accuracy	V _{ACC_VGH}	no load, V _{VGH} =18V	-3		+3	%
Under Voltage Protection	V _{UVP_VGH}	V _{VGH} falling under Target Percentage	-	80	-	%
V _{GH} Fault Condition		UVP continued occur	-	50	-	ms
Current Limit	I _{LIM_VGH}		-	1.08	-	A
Maximum Driving Ability		V _{GH} =18V, f _{SIBO} =715kHz	10	-	-	mA
SIBO Converter – V_{GL2}						
Operating Frequency	f _{SIBO}	500/600/715/800/933/1000 /1225/1430kHz		F _{OSC_LXN}		kHz
Frequency Accuracy		f _{SIBO} =715kHz	-5		+5	%
		Others	-10		+10	%
Output Voltage Range	V _{VGL2}	0.2V/step	-18	-	-5.4	V
Output Voltage Accuracy	V _{ACC_VGL2}	no load, V _{VGL2} =-12V	-3		+3	%
Under Voltage Protection	V _{UVP_VGL2}		-	NVDD-1	-	V

VGL2 Fault Condition		UVP continued occur	-	50	-	ms
Current Limit	I _{LIM_VGL2}		-	750/1500	-	mA
Maximum Driving Abilit.y		VIN=3.3V, VGL2=-12V, f _{SIBO} =715kHz	10	-	-	mA
Voltage Detector-RESET1 and RESET2						
Detection Voltage	V _{DET}	100mV/step	2	-	2.7	V
Detecting Voltage Accuracy	V _{ACC_DET}	No load	-3		+3	%
Programmable V _{COM} and output buffer – VCOM						
VCOM Output Voltage	V _{COM}	0.01V/step	-3.6	-	2.02	V
Output Voltage Accuracy	V _{ACC_VCOM}	No load	-15	-	15	mV
Output Voltage Ability		each step, I _{OUT} = +/-10mA	-	15	-	mV
Short-Circuit Current		each step, maximum source/sink current	-	200	-	mA
Programmable gamma voltage and output buffer – GMA1 and GMA2						
GMA1 Output Voltage	V _{GMA1}	0.02V/step	PVDD -1.26	-	PVDD -0.1	V
GMA2 Output Voltage	V _{GMA2}	0.02V/step	NVDD +0.1	-	NVDD +1.26	V
Output Voltage accuracy	V _{ACC_GMA}	No load	-20	-	20	mV
Short-Circuit Current			-	20	-	mA
CTRL PIN						
Threshold voltage	V _{IH}		1	-	-	V
	V _{IL}		-	-	0.5	
I2C INTERFACE						
Low Level Input Voltage	V _{IL_I2C}	VIN=2.5V to 5.0V	-	-	0.5	V
High Level Input Voltage	V _{IH_I2C}	VIN=2.5V to 5.0V	1	-	-	V
EEPROM Write Time	T _{WRITE}		-	-	200	ms
Number of Guaranteed EEPROM Write Cycles	N _{WRITE}		-	-	100	Cycle s

Power On/Off Sequence



[Figure 1] Power Sequence

DETAILED DESCRIPTION

1. Under Voltage Lock Out (VIN UVLO)

To ensure that the input voltage (VIN) is high enough for reliable operation, the LX52042C includes an under voltage lockout (UVLO) circuit. The UVLO threshold at the VIN pin is 2.4V (typ.) rising and 1.9V (typ.) falling. Once the input voltage exceeds the UVLO rising threshold, the controller enables the reference block. When the input voltage falls below the UVLO falling threshold, the controller turns off the reference and all the IC's outputs.

2. Programmable Soft Start

The PVDD/NVDD boost, VGH/VGL2 SIBO, VCORE/VIO buck, and VGL1 LDO have programmable soft-start functions. The soft start time is controlled by register setting value

3. Programmable Delay Time

The PVDD/NVDD boost, VGH/VGL2 SIBO, VCORE/VIO buck, and VGL1 LDO have a programmable delay time function. The delay time is controlled by register setting value

4. Over Temperature Protection (OTP)

Thermal protection is implemented to prevent damage due to excessive heat and power dissipation. Typically, the thermal protection threshold is 150°C. If this temperature is reached, the device goes into shut-down.

5. Over Voltage Protection (OVP)

The converters have over voltage protections for maximum safety. The over voltage protection is implemented to protect the LX52042C and / or peripheral IC in case the output voltage exceeds the normal operation range. Once the over voltage event is detected, the protection circuits stop switching until the output voltage falls below the over voltage threshold, and then the converter continues normal operation

6. Over Current Protection (OCP)

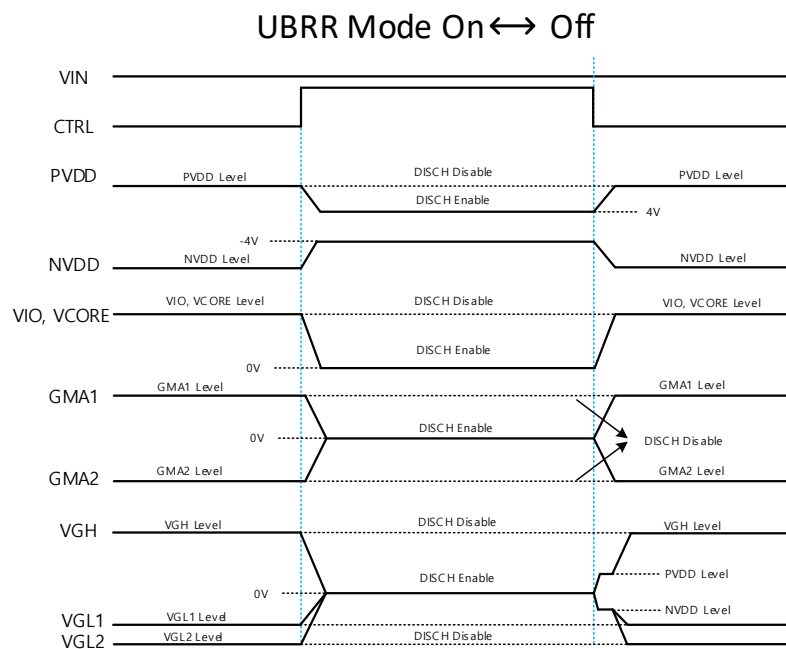
The over current protections of the LX52042C converters set a limitation to the inductor's input peak current. If the input current exceeds the maximum peak current level (ex: The PVDD boost's maximum level is 2.5A), the input current is clamped at the maximum peak current level by an internal over current protection circuitry.

7. VGH Temperature Compensation

As the temperature is lower, TFT's turn-on threshold is higher. So, the VGH boost converter has a low temperature compensation function to increase VGH, the TFT's turn-on voltage, in low-temperature conditions.

8. Protection Table

Protection Category	Item	Condition (Enable Timing)	Description
PVDD	UVP	ON: >PVDD's 80% OFF: <PVDD's 80%	1) UVP is enabled after PVDD soft-start 2) Maintain for 50ms, all converters are disabled
	OCP	ON: >OCP Level	1) OCP is enabled when PVDD enables 2) PVDD switch off
	OVP	ON: > 7.79V OFF: <7.79V	1) OVP is enabled when PVDD enables 2) PVDD switch off
NVDD	UVP	ON: > NVDD 's 80% OFF: < NVDD 's 80%	1) UVP is enabled after NVDD soft-start 2) Maintain for 50ms, all converters are disabled
	OCP	ON: >OCP Level	1) OCP is enabled when NVDD enables 2) NVDD switch off
	OVP	ON: < -7.8V OFF: > -7.8V	1) OVP is enabled when NVDD enables 2) NVDD switch off
VGH	UVP	ON: >VGH's 80% OFF: <VGH's 80%	1) UVP is enabled after VGH soft-start 2) Maintain for 50ms, all converters are disabled
	OCP	ON: >OCP Level	1) OCP is enabled when VGH enables 2) VGH switch off
	OVP	ON: > 30.5V OFF: <30.5V	1) OVP is enabled when VGH enables 2) VGH switch off
VGL2	UVP	ON: > NVDD-1V OFF: < NVDD-1V	1) UVP is enabled after VGL2 soft-start 2) Maintain for 50ms, all converters are disabled
	OCP	ON: > OCP Level	1) OCP is enabled when VGL2 enables 2) VGH switch off
	OVP	ON: < -19.8V OFF: > -19.8V	1) OVP is enabled when VGL2 enables 2) VGL2 switch off
VIO & VCORE	UVP	ON: >VOUT's 80% OFF: <VOUT's 80%	1) UVP is enabled after Buck soft-start 2) Maintain for 60ms, all converters are disabled
	OCP	ON: >OCP Level	1) OCP is enabled when Buck enables 2) Buck switch off
VGL1	UVP	ON: > NVDD-1V OFF: < NVDD-1V	1) UVP is enabled after VGL1 soft-start 2) Maintain for 50ms, all converters are disabled



[Figure 2] Power Sequence @UBRR Mode

Note

1) Each IP's enable condition in the UBRR mode is determined by the UBRR CTRL option, either enable or disable. The discharge option of LX52042C is determined by the channel setting option (02h) during the UBRR operation.

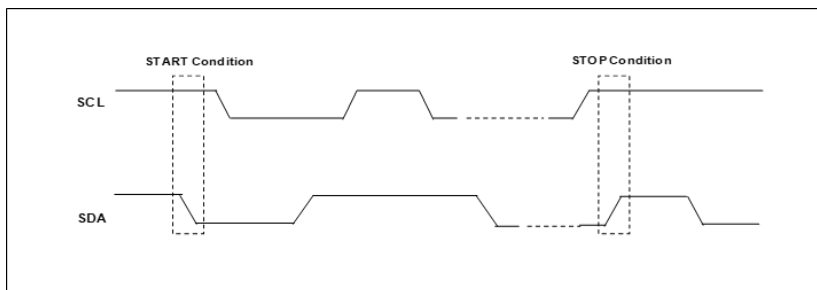
I2C Control

1. Start Condition

At idle bus state, both SDA and SCL are high level. To setup the START condition, please change the SDA state to low from high while SCL is high state.

2. Stop Condition

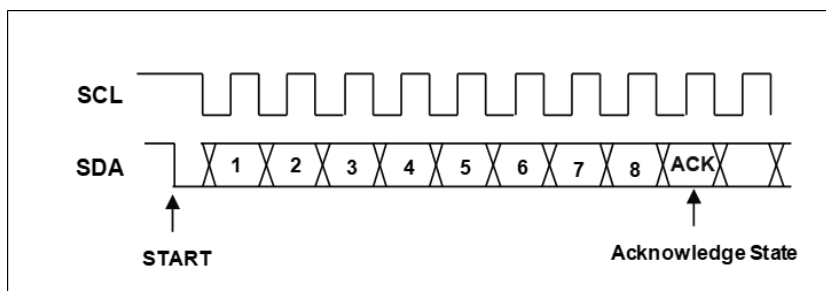
To setup the STOP condition, please change the SDA state to high from low while SCL is high state.



[Figure 3] Start & Stop Condition Timing Diagram of the I2C Interface

3. Bus Acknowledge

The Acknowledge (ACK) signal is generated after the transfer for each byte is finished. If the SDA bus goes to low at ACK time, data transfer is passed. If the SDA bus goes to high, data transfer is failed. In the case of transfer fail, you can't write the data to the register.



[Figure 4] 1-byte Timing Diagram of the I2C Interface

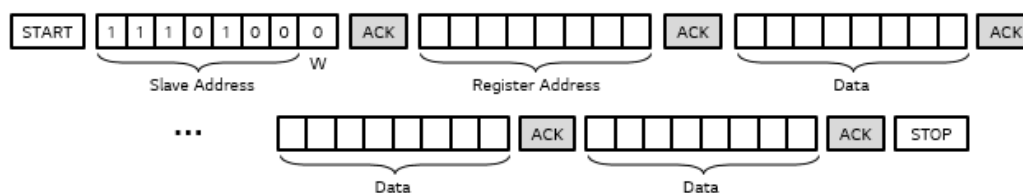
4. I2C Protocol for PMIC

Description	D7	D6	D5	D4	D3	D2	D1	D0	Default
Slave address (74h, 7-bit) Read=1/Write=0	1	1	1	0	1	0	0	R/W	E8h (8-bit)

a) Write Single Byte to Register



b) Write Multiple Bytes to Register



c) Write all DAC Registers to EEPROM

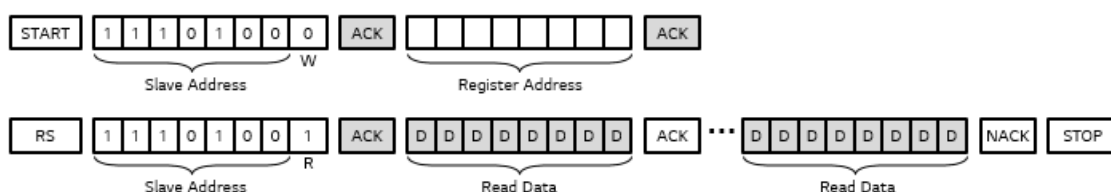


d) Read Data from Register

Step 1: Command for reading Register



Step 2: Read Command

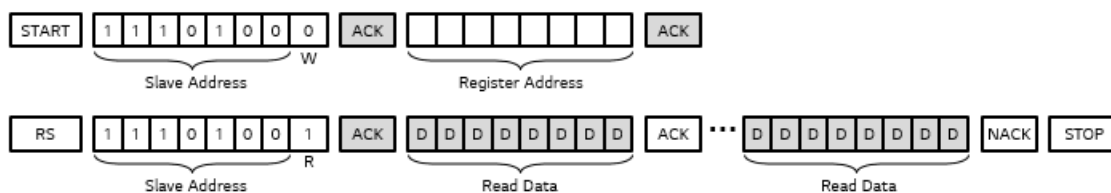


e) Read Data from EEPROM

Step 1: Command for reading EEPROM



Step 2: Read Command



5. I2C Protocol for D_VCOM

Description	D7	D6	D5	D4	D3	D2	D1	D0	Default
DVCOM Slave address (28h, 7-bit) Read=1/Write=0	0	1	0	1	0	0	0	R/W	50h (8-bit)
PVCOM HData (00h)	VCOM [7:0]								80h
PVCOM LData (01h)	-	-	-	-	-	VCOM_LSB [2:0]			00h

PVCOM_SEL (02h)	PVCOM SEL	-	-	-	-	-	-	-	00h
--------------------	--------------	---	---	---	---	---	---	---	-----

During Write command, PVCOM_SEL=0: write data to Register & EEPROM;
PVCOM_SEL=1: write data to Register.

During Read command, PVCOM_SEL=0: read data from EEPROM
PVCOM_SEL=1: read data from Register

a) Write Data to Register & EEPROM

Step 1: set PVCOM_SEL = 0



Step 2: Write Command



b) Write Data to Register

Step 1: set PVCOM_SEL = 1



Step 2: Write Command

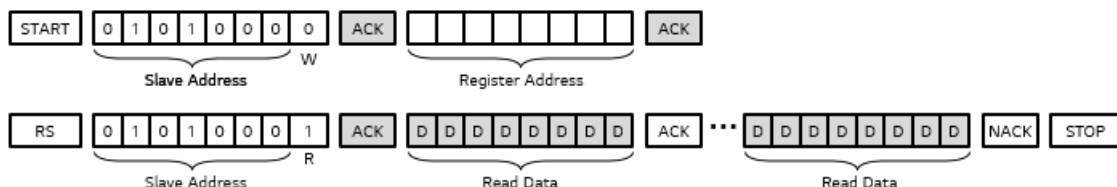


c) Read Data from EEPROM

Step 1: set PVCOM_SEL = 0



Step 2: Read Command

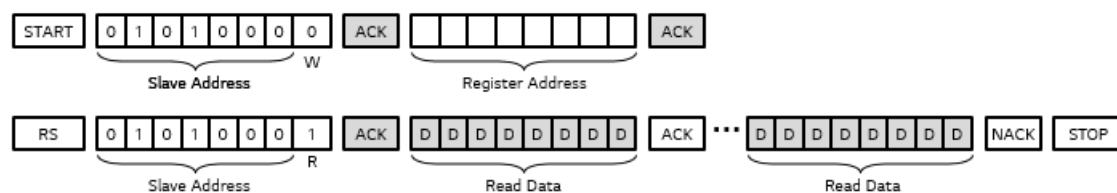


d) Read Data from Register

Step 1: set PVCOM_SEL = 1



Step 2: Read Command



6. I2C Interface Characteristics

$V_{IN}=V_{INP}=V_{INN}=3.3V$, $V_{IN_SIBO}=PVDD=5V$, $NVDD=-5V$, $V_{GH}=18V$, $V_{GL2}=-12V$, $V_{GL1}=-9V$, $V_{CORE}=1.2V$, $V_{IO}=1.8V$, $GMA1=4.8V$, $GMA2=-4.8V$, $PGND=AGND=0V$, $Freq=715kHz$, All Grounds are 0V, $T_A=25^{\circ}C$ (unless otherwise noted)

I ² C Interface ¹⁾²⁾						
PARAMETER	SYMBOL	CONDITION	MIN	TYP	MAX	UNIT
Logic Input Threshold High	V_{IH_I2C}	SCL, SDA @ $V_{IN}=3.3V$	1.5		V_{IN}	V
Logic Input Threshold Low	V_{IL_I2C}	SCL, SDA @ $V_{IN}=3.3V$			0.5	V
SDA Logic Output Low Voltage	V_{OL_I2C}	SDA @sink 3mA	0		0.4	V
Logic Input Leakage Current	I_{LK_I2C}	SCL, SDA	-1		1	μA
SDA and SCL Input Capacitance	C_{IN_I2C}	SCL, SDA		5	10	pF
SCL Clock Frequency	f_{CLK_SCL}				400	kHz
SCL High Period	T_{HP_SCL}		600			ns
SCL Low Period	T_{LP_SCL}		1300			ns
SDA and SCL Rise Time	T_{R_I2C}	C_b = Total cap of one bus in pF		$20+10C_b$	300	ns
SDA and SCL Fall Time	T_{F_I2C}	C_b = Total cap of one bus in pF		$20+10C_b$	300	ns
START Condition Hold/Setup Time	T_{STA_I2C}	10% of SDA to 90% of SDA	600			ns
Data Input Hold Time	T_{HD_SDA}		0		900	ns
Data Input Setup Time	T_{SU_SDA}		100			ns
STOP Condition Setup Time	T_{STP_I2C}		600			ns
Bus Free Time	T_{BUF_I2C}		1300			ns
SDA Capacitive Loading	C_{L_SDA}				400	pF

PVCOM Memory Write Time					50-	ms
PMIC Memory Write Time					200-	ms
Memory Endurance					100	cycle

Note

- 1) Guaranteed by design, characterization and correlation with process controls. Not fully tested in production.
- 2) Designed and simulated according to I2C specifications except general call support.

Register Map

Item	Address	Register	Default	Default Value	Range	Resolution	Bit
Channel Setting	00h	EN_PVDD	FFh	Enable			1
		EN_VGH		Enable			1
		EN_VGL2		Enable			1
		EN_VIO		Enable			1
		EN_VCORE		Enable			1
		EN_VGL1		Enable			1
		EN_VCOM		Enable			1
	01h	EN_GMA1	8Bh	Enable			1
		EN_GMA2		Enable			1
		EN_RESET1		Disable			1
		U_NVDD_4p7uH		Enable			1
		EN_RESET2		Disable			1
		VGH_High_Resolution		Enable			1
	02h	PVDD_DISCH	00h	Disable			1
		NVDD_DISCH		Disable			1
		VGH_DISCH		Disable			1
		VGL2_DISCH		Disable			1
		VCORE_DISCH		Disable			1
		VIO_DISCH		Disable			1
		VGL1_DISCH		Disable			1
		VCOM_DISCH		Disable			1
PVDD Voltage	03h	PVDD[5:0]	14h	5.0V	4.0V~7.0V	0.05V	6
NVDD Voltage	04h	NVDD[5:0]	14h	-5.0V	-7.15V~-4.0V	0.05V	6
VGH Voltage	05h	VGH[4:0]	0Bh	8.2V	6.0V~12.0V	0.2V	5
VGL2 Voltage	06h	VGL2[5:0]	21h	18.0V	12.5V~28.0V	0.5V	5
VCORE Voltage	07h	VCORE[5:0]	14h	-12V	-18.0V~-5.4V	0.2V	6
VIO Voltage	08h	VIO[4:0]	10h	0.9V	0.8V~2.06V	0.02V	6
VGL1 Voltage	09h	VGL1[5:0]	12h	1.8V	1.0V~2.55V	0.05V	5
VCOM Voltage	0Ah	VCOM[7:0]	80h	-9.0V	-18.0V~-5.4V	0.2V	6
VCOM_LSB Voltage	0Bh	VCOM_LSB[2:0]	00h	VCOM_MIN +1.28V	0V~2.55V (from VCOM_MIN)	0.01V	8
VRESET Voltage	0Ch	VRESET2[2:0]	00h	0V	0mV~8.75mV	1.25mV	3
		VRESET1[2:0]		2.0V	2.0V~2.7V	0.1V	3
GMA1 Voltage	0Dh	GMA1[5:0]	0Ah	2.0V	2.0V~2.7V	0.1V	3
GMA2 Voltage	0Eh	GMA2[5:0]	0Ah	PVDD-0.2V	(PVDD-0.1)~(PVDD-1.26V)	0.02V	6
PVDD Boost Control	0Fh	LXP_OCP[1:0]	99h	NVDD+0.2V	(NVDD+0.1)~(NVDD+1.26V)	0.02V	6
		LXP_SR[2:0]		2.5A	1.5~3A	0.5A	2
		LXP_FREQ[2:0]		LV4	LV1~7	-	3
	10h	PVDD_SS[2:0]	01h	715kHz	600~1225kHz	-	3
NVDD Boost Control	11h	PVDD_DLY[3:0]		2ms	2~16ms	2ms	3
		NVDD_SS[2:0]	01h	5ms	0~45ms	5ms	4
	1Ch	NVDD_DLY[3:0]		2ms	2~16ms	2ms	3
		LXN_SR[6:5]	48h	5ms	0~45ms	5ms	4
VGX SIBO Control	12h	LXN_OCP[4:3]		-	-	-	2
		LXH/L_SR[1:0]	12h	2.0A	1~2.5A	0.5A	2
		LX_H/L_FREQ[2:0]		-	-	-	2
	13h	SIBO_FREQ	03h	715kHz	50~1430kHz	-	3
		VGH_SS[1:0]		Default	Default / Half	-	1
		VGH_DLY[2:0]		4ms	4~16ms	4ms	2
	14h	VGL2_SS[2:0]	03h	15ms	0~45ms	5ms	4
		VGL2_DLY[3:0]		4ms	4~16ms	2ms	3
VCORE Buck Control	15h	VCORE_SR[1:0]	11h	15ms	0~45ms	5ms	4
		VCORE_FREQ[2:0]		-	-	-	2
	16h	VCORE_SS[1:0]	00h	715kHz	600~1225kHz	-	3
		VCORE_DLY[3:0]		2ms	2~8ms	2ms	2
VIO Buck Configuration	17h	VIO_SR[1:0]	11h	0ms	0~45ms	5ms	4
		VIO_FREQ[2:0]		-	-	-	2
	18h	VIO_SS[1:0]	00h	715kHz	600~1225kHz	-	3
		VIO_DLY[3:0]		2ms	2~8ms	2ms	2
RESET Delay	19h	RESET2_DLY[3:0]	10h	0ms	0~45ms	5ms	4
		RESET1_DLY[3:0]		5ms	0~45ms	5ms	4
VCOM	1Bh	VCOM_OFF_SEL	02h	-	-	-	1

Control		VCOM_DLY[4:0]		10ms	0~155ms	5ms	5
VCOM_MIN	1Dh	VCOM_MIN[4:0]	0Fh	-1.65V	-3.6~0.75V	0.15V	5
RCOMP	1Fh	VGL2_RCOMP[1:0]	AAh	-	-	-	2
		VGH_RCOMP[1:0]		-	-	-	2
		NVDD_RCOMP[1:0]		-	-	-	2
		PVDD_RCOMP[1:0]		-	-	-	2
	20h	VCORE_RCOMP[1:0]	05h	-	-	-	2
		VIO_RCOMP[1:0]		-	-	-	2
Option	21h	BUCK_BUF_RD[1:0]	AFh	-	-	-	2
		VGH_LT_SEL[1:0]		6V	2~8V	2V	2
		OTP_EN		Enable	-	-	1
		UVP_EN		Enable	-	-	1
		ABN_EN[1:0]		-	-	-	2
		SIBO_EN		-	-	-	1
	22h	VG_PHASE[1:0]	61h	50%	40~55%	-	1
		VGH_preSS_5ms		1ms	1~5ms	-	2
		VGH_PRECHG		Disable	-	-	1
		VGL2_OCPH		0.75A	0.75~1.5A	0.75A	1
		VCOM_BIAS		-	-	-	1
UBRR Control1	23h	EN_VGL1_UBRR3	0Fh	Enable	Disable / Enable	-	1
		EN_VGL2_UBRR3		Enable	Disable / Enable	-	1
		EN_NVDD_UBRR3		Enable	Disable / Enable	-	1
		EN_PVDD_UBRR3		Enable	Disable / Enable	-	1
UBRR Control2	24h	UBRR_IN4	00h	Close	Open / Close	-	1
		UBRR_IN3		Close	Open / Close	-	1
		UBRR_IN2		Close	Open / Close	-	1
		UBRR_IN1		Close	Open / Close	-	1
UBRR Control3	25h	iCOMP_Use[1:0]	06h	-	-	-	2
		iCTRL_Use[1:0]		-	-	-	2
UBRR Control4	26h	EN_VCOM_UBRR1	FFh	Enable	Disable / Enable	-	1
		EN_VGL1_UBRR1		Enable	Disable / Enable	-	1
		EN_VCORE_UBRR1		Enable	Disable / Enable	-	1
		EN_VIO_UBRR1		Enable	Disable / Enable	-	1
		EN_VGL2_UBRR1		Enable	Disable / Enable	-	1
		EN_VGH_UBRR1		Enable	Disable / Enable	-	1
		EN_NVDD_UBRR1		Enable	Disable / Enable	-	1
		EN_PVDD_UBRR1		Enable	Disable / Enable	-	1
UBRR Control5	27h	EN_VCORE_UBRR3	3Fh	Disable	Disable / Enable	-	1
		EN_VIO_UBRR3		Disable	Disable / Enable	-	1
		EN_RESET1_UBRR1		Enable	Disable / Enable	-	1
		EN_RESET2_UBRR1		Enable	Disable / Enable	-	1
		EN_GMA_UBRR1		Enable	Disable / Enable	-	1
		UBRR1_DISCH_EN		Enable	Disable / Enable	-	1
		PVDD_DOWN1		Enable	Disable / Enable	-	1
		NVDD_DOWN1		Enable	Disable / Enable	-	1
UBRR Control6	28h	PVDD_DL3[3:0]	00h	4.0V	4.0V~5.5V	0.1V	4
		NVDD_DL3[3:0]		4.0V	-4.0V~-5.5V	-0.1V	4
UBRR Control7	29h	EN_VCOM_UBRR2	FFh	Enable	Disable / Enable	-	1
		EN_VGL1_UBRR2		Enable	Disable / Enable	-	1
		EN_VCORE_UBRR2		Enable	Disable / Enable	-	1
		EN_VIO_UBRR2		Enable	Disable / Enable	-	1
		EN_VGL2_UBRR2		Enable	Disable / Enable	-	1
		EN_VGH_UBRR2		Enable	Disable / Enable	-	1
		EN_NVDD_UBRR2		Enable	Disable / Enable	-	1
		EN_PVDD_UBRR2		Enable	Disable / Enable	-	1
UBRR Control8	2Ah	NVDD_DOWN3	FFh	Enable	Disable / Enable	-	1
		PVDD_DOWN3		Enable	Disable / Enable	-	1
		EN_RESET1_UBRR2		Enable	Disable / Enable	-	1
		EN_RESET2_UBRR2		Enable	Disable / Enable	-	1
		EN_GMA_UBRR2		Enable	Disable / Enable	-	1
		UBRR2_DISCH_EN		Enable	Disable / Enable	-	1
		PVDD_DOWN2		Enable	Disable / Enable	-	1
		NVDD_DOWN2		Enable	Disable / Enable	-	1
Control	FEh	WED_VCOM	00h	-	-	-	1
	FFh	WED_ALL	00h	Disable	-	-	1
		RED		Disable	-	-	1

※ Not fully tested in all register parameters. Please refer to electrical characteristic tables.

CH_SET1 (00h) – IP Enable

7		6		5		4		3		2		1		0	
EN_VCOM		EN_VGL1		EN_VCORE		EN_VIO		EN_VGL2		EN_VGH		Reserved		EN_PVDD	
00h	Disable	00h	Disable	00h	Disable	00h	Disable	00h	Disable	00h	Disable			00h	Disable
01h	<u>Enable</u>	01h	<u>Enable</u>	01h	<u>Enable</u>	01h	<u>Enable</u>	01h	<u>Enable</u>	01h	<u>Enable</u>			01h	<u>Enable</u>

CH_SET2 (01h) – IP Enable

7		6		5		4		3		2		1		0	
VGH_HighResolution		Reserved		Reserved		EN_RESET2		U_NVDD_4p7uH		EN_RESET1		EN_GMA2		EN_GMA1	
00h	Disable					00h	Disable	00h	Disable	00h	Disable	00h	Disable	00h	Disable
01h	<u>Enable</u>					01h	<u>Enable</u>	01h	<u>Enable</u>	01h	<u>Enable</u>	01h	<u>Enable</u>	01h	<u>Enable</u>

CH_SET 3(02h) – IP Enable

7		6		5		4		3		2		1		0	
VCOM_DISCH		VGL1_DISCH		VCORE_DISCH		VIO_DISCH		VGL2_DISCH		VGH_DISCH		NVDD_DISCH		PVDD_DISCH	
00h	<u>Disable</u>	00h	<u>Disable</u>	00h	<u>Disable</u>	00h	<u>Disable</u>	00h	<u>Disable</u>	00h	<u>Disable</u>	00h	<u>Disable</u>	00h	<u>Disable</u>
01h	<u>Enable</u>	01h	<u>Enable</u>	01h	<u>Enable</u>	01h	<u>Enable</u>	01h	<u>Enable</u>	01h	<u>Enable</u>	01h	<u>Enable</u>	01h	<u>Enable</u>

VPVDD (03h) – PVDD voltage

7		6		5		4		3		2		1		0	
Reserved				V _{PVDD}											
00h V _{PVDD} =4.00V				10h V _{PVDD} =4.80V				20h V _{PVDD} =5.60V				30h V _{PVDD} =6.40V			
01h V _{PVDD} =4.05V				11h V _{PVDD} =4.85V				21h V _{PVDD} =5.65V				31h V _{PVDD} =6.45V			
02h V _{PVDD} =4.10V				12h V _{PVDD} =4.90V				22h V _{PVDD} =5.70V				32h V _{PVDD} =6.50V			
03h V _{PVDD} =4.15V				13h V _{PVDD} =4.95V				23h V _{PVDD} =5.75V				33h V _{PVDD} =6.55V			
04h V _{PVDD} =4.20V				14h <u>V_{PVDD}=5.00V</u>				24h V _{PVDD} =5.80V				34h V _{PVDD} =6.60V			
05h V _{PVDD} =4.25V				15h V _{PVDD} =5.05V				25h V _{PVDD} =5.85V				35h V _{PVDD} =6.65V			
06h V _{PVDD} =4.30V				16h V _{PVDD} =5.10V				26h V _{PVDD} =5.90V				36h V _{PVDD} =6.70V			
07h V _{PVDD} =4.35V				17h V _{PVDD} =5.15V				27h V _{PVDD} =5.95V				37h V _{PVDD} =6.75V			
08h V _{PVDD} =4.40V				18h V _{PVDD} =5.20V				28h V _{PVDD} =6.00V				38h V _{PVDD} =6.80V			
09h V _{PVDD} =4.45V				19h V _{PVDD} =5.25V				29h V _{PVDD} =6.05V				39h V _{PVDD} =6.85V			
0Ah V _{PVDD} =4.50V				1Ah V _{PVDD} =5.30V				2Ah V _{PVDD} =6.10V				3Ah V _{PVDD} =6.90V			
0Bh V _{PVDD} =4.55V				1Bh V _{PVDD} =5.35V				2Bh V _{PVDD} =6.15V				3Bh V _{PVDD} =6.95V			
0Ch V _{PVDD} =4.60V				1Ch V _{PVDD} =5.40V				2Ch V _{PVDD} =6.20V				3Ch V _{PVDD} =7.00V			
0Dh V _{PVDD} =4.65V				1Dh V _{PVDD} =5.45V				2Dh V _{PVDD} =6.25V				3Dh V _{PVDD} =7.00V			
0Eh V _{PVDD} =4.70V				1Eh V _{PVDD} =5.50V				2Eh V _{PVDD} =6.30V				3Eh V _{PVDD} =7.00V			
0Fh V _{PVDD} =4.75V				1Fh V _{PVDD} =5.55V				2Fh V _{PVDD} =6.35V				3Fh V _{PVDD} =7.00V			

VNVDD (04h) – NVDD voltage

7	6	5	4	3	2	1	0
Reserved		V _{NVDD}					
00h	V _{NVDD} = -4.00V	10h	V _{NVDD} = -4.80V	20h	V _{NVDD} = -5.60V	30h	V _{NVDD} = -6.40V
01h	V _{NVDD} = -4.05V	11h	V _{NVDD} = -4.85V	21h	V _{NVDD} = -5.65V	31h	V _{NVDD} = -6.45V
02h	V _{NVDD} = -4.10V	12h	V _{NVDD} = -4.90V	22h	V _{NVDD} = -5.70V	32h	V _{NVDD} = -6.50V
03h	V _{NVDD} = -4.15V	13h	V _{NVDD} = -4.95V	23h	V _{NVDD} = -5.75V	33h	V _{NVDD} = -6.55V
04h	V _{NVDD} = -4.20V	14h	V _{NVDD} = -5.00V	24h	V _{NVDD} = -5.80V	34h	V _{NVDD} = -6.60V
05h	V _{NVDD} = -4.25V	15h	V _{NVDD} = -5.05V	25h	V _{NVDD} = -5.85V	35h	V _{NVDD} = -6.65V
06h	V _{NVDD} = -4.30V	16h	V _{NVDD} = -5.10V	26h	V _{NVDD} = -5.90V	36h	V _{NVDD} = -6.70V
07h	V _{NVDD} = -4.35V	17h	V _{NVDD} = -5.15V	27h	V _{NVDD} = -5.95V	37h	V _{NVDD} = -6.75V
08h	V _{NVDD} = -4.40V	18h	V _{NVDD} = -5.20V	28h	V _{NVDD} = -6.00V	38h	V _{NVDD} = -6.80V
09h	V _{NVDD} = -4.45V	19h	V _{NVDD} = -5.25V	29h	V _{NVDD} = -6.05V	39h	V _{NVDD} = -6.85V
0Ah	V _{NVDD} = -4.50V	1Ah	V _{NVDD} = -5.30V	2Ah	V _{NVDD} = -6.10V	3Ah	V _{NVDD} = -6.90V
0Bh	V _{NVDD} = -4.55V	1Bh	V _{NVDD} = -5.35V	2Bh	V _{NVDD} = -6.15V	3Bh	V _{NVDD} = -6.95V
0Ch	V _{NVDD} = -4.60V	1Ch	V _{NVDD} = -5.40V	2Ch	V _{NVDD} = -6.20V	3Ch	V _{NVDD} = -7.00V
0Dh	V _{NVDD} = -4.65V	1Dh	V _{NVDD} = -5.45V	2Dh	V _{NVDD} = -6.25V	3Dh	V _{NVDD} = -7.05V
0Eh	V _{NVDD} = -4.70V	1Eh	V _{NVDD} = -5.50V	2Eh	V _{NVDD} = -6.30V	3Eh	V _{NVDD} = -7.10V
0Fh	V _{NVDD} = -4.75V	1Fh	V _{NVDD} = -5.55V	2Fh	V _{NVDD} = -6.35V	3Fh	V _{NVDD} = -7.15V

VGH (05h) – VGH voltage

7	6	5	4	3	2	1	0
Reserved			V _{GH}				
When VGH_HighResolution = 1							
00h	V _{GH} =12.5V	08h	V _{GH} =16.5V	10h	V _{GH} =20.5V	18h	V _{GH} =24.5V
01h	V _{GH} =13.0V	09h	V _{GH} =17.0V	11h	V _{GH} =21.0V	19h	V _{GH} =25.0V
02h	V _{GH} =13.5V	0Ah	V _{GH} =17.5V	12h	V _{GH} =21.5V	1Ah	V _{GH} =25.5V
03h	V _{GH} =14.0V	0Bh	<u>V_{GH}=18.0V</u>	13h	V _{GH} =22.0V	1Bh	V _{GH} =26.0V
04h	V _{GH} =14.5V	0Ch	V _{GH} =18.5V	14h	V _{GH} =22.5V	1Ch	V _{GH} =26.5V
05h	V _{GH} =15.0V	0Dh	V _{GH} =19.0V	15h	V _{GH} =23.0V	1Dh	V _{GH} =27.0V
06h	V _{GH} =15.5V	0Eh	V _{GH} =19.5V	16h	V _{GH} =23.5V	1Eh	V _{GH} =27.5V
07h	V _{GH} =16.0V	0Fh	V _{GH} =20.0V	17h	V _{GH} =24.0V	1Fh	V _{GH} =28.0V
When VGH_HighResolution = 0							
00h	V _{GH} =6.0V	08h	V _{GH} =7.6V	10h	V _{GH} =9.2V	18h	V _{GH} =10.8V
01h	V _{GH} =6.2V	09h	V _{GH} =7.8V	11h	V _{GH} =9.4V	19h	V _{GH} =11.0V
02h	V _{GH} =6.4V	0Ah	V _{GH} =8.0V	12h	V _{GH} =9.6V	1Ah	V _{GH} =11.2V
03h	V _{GH} =6.6V	0Bh	V _{GH} =8.2V	13h	V _{GH} =9.8V	1Bh	V _{GH} =11.4V
04h	V _{GH} =6.8V	0Ch	V _{GH} =8.4V	14h	V _{GH} =10.0V	1Ch	V _{GH} =11.6V
05h	V _{GH} =7.0V	0Dh	V _{GH} =8.6V	15h	V _{GH} =10.2V	1Dh	V _{GH} =11.8V
06h	V _{GH} =7.2V	0Eh	V _{GH} =8.8V	16h	V _{GH} =10.4V	1Eh	V _{GH} =12.0V
07h	V _{GH} =7.4V	0Fh	V _{GH} =9.0V	17h	V _{GH} =10.6V	1Fh	V _{GH} =12.0V

VGL2 (06h) – VGL2 voltage

7	6	5	4	3	2	1	0
Reserved		V _{GL2}					
00h V _{GL2} =-5.4V		10h V _{GL2} =-8.6V		20h V _{GL2} =-11.8V		30h V _{GL2} =-15.0V	
01h V _{GL2} =-5.6V		11h V _{GL2} =-8.8V		21h V _{GL2} =-12.0V		31h V _{GL2} =-15.2V	
02h V _{GL2} =-5.8V		12h V _{GL2} =-9.0V		22h V _{GL2} =-12.2V		32h V _{GL2} =-15.4V	
03h V _{GL2} =-6.0V		13h V _{GL2} =-9.2V		23h V _{GL2} =-12.4V		33h V _{GL2} =-15.6V	
04h V _{GL2} =-6.2V		14h V _{GL2} =-9.4V		24h V _{GL2} =-12.6V		34h V _{GL2} =-15.8V	
05h V _{GL2} =-6.4V		15h V _{GL2} =-9.6V		25h V _{GL2} =-12.8V		35h V _{GL2} =-16.0V	
06h V _{GL2} =-6.6V		16h V _{GL2} =-9.8V		26h V _{GL2} =-13.0V		36h V _{GL2} =-16.2V	
07h V _{GL2} =-6.8V		17h V _{GL2} =-10.0V		27h V _{GL2} =-13.2V		37h V _{GL2} =-16.4V	
08h V _{GL2} =-7.0V		18h V _{GL2} =-10.2V		28h V _{GL2} =-13.4V		38h V _{GL2} =-16.6V	
09h V _{GL2} =-7.2V		19h V _{GL2} =-10.4V		29h V _{GL2} =-13.6V		39h V _{GL2} =-16.8V	
0Ah V _{GL2} =-7.4V		1Ah V _{GL2} =-10.6V		2Ah V _{GL2} =-13.8V		3Ah V _{GL2} =-17.0V	
0Bh V _{GL2} =-7.6V		1Bh V _{GL2} =-10.8V		2Bh V _{GL2} =-14.0V		3Bh V _{GL2} =-17.2V	
0Ch V _{GL2} =-7.8V		1Ch V _{GL2} =-11.0V		2Ch V _{GL2} =-14.2V		3Ch V _{GL2} =-17.4V	
0Dh V _{GL2} =-8.0V		1Dh V _{GL2} =-11.2V		2Dh V _{GL2} =-14.4V		3Dh V _{GL2} =-17.6V	
0Eh V _{GL2} =-8.2V		1Eh V _{GL2} =-11.4V		2Eh V _{GL2} =-14.6V		3Eh V _{GL2} =-17.8V	
0Fh V _{GL2} =-8.4V		1Fh V _{GL2} =-11.6V		2Fh V _{GL2} =-14.8V		3Fh V _{GL2} =-18.0V	

VCORE (07h) – VCORE voltage

7	6	5	4	3	2	1	0
Reserved		V _{CORE}					
00h V _{CORE} =0.8V		10h V _{CORE} =1.12V		20h V _{CORE} =1.44V		30h V _{CORE} =1.76V	
01h V _{CORE} =0.82V		11h V _{CORE} =1.14V		21h V _{CORE} =1.46V		31h V _{CORE} =1.78V	
02h V _{CORE} =0.84V		12h V _{CORE} =1.16V		22h V _{CORE} =1.48V		32h V _{CORE} =1.80V	
03h V _{CORE} =0.86V		13h V _{CORE} =1.18V		23h V _{CORE} =1.50V		33h V _{CORE} =1.82V	
04h V _{CORE} =0.88V		14h V _{CORE} =1.20V		24h V _{CORE} =1.52V		34h V _{CORE} =1.84V	
05h V _{CORE} =0.90V		15h V _{CORE} =1.22V		25h V _{CORE} =1.54V		35h V _{CORE} =1.86V	
06h V _{CORE} =0.92V		16h V _{CORE} =1.24V		26h V _{CORE} =1.56V		36h V _{CORE} =1.88V	
07h V _{CORE} =0.94V		17h V _{CORE} =1.26V		27h V _{CORE} =1.58V		37h V _{CORE} =1.90V	
08h V _{CORE} =0.96V		18h V _{CORE} =1.28V		28h V _{CORE} =1.60V		38h V _{CORE} =1.92V	
09h V _{CORE} =0.98V		19h V _{CORE} =1.30V		29h V _{CORE} =1.62V		39h V _{CORE} =1.94V	
0Ah V _{CORE} =1.00V		1Ah V _{CORE} =1.32V		2Ah V _{CORE} =1.64V		3Ah V _{CORE} =1.96V	
0Bh V _{CORE} =1.02V		1Bh V _{CORE} =1.34V		2Bh V _{CORE} =1.66V		3Bh V _{CORE} =1.98V	
0Ch V _{CORE} =1.04V		1Ch V _{CORE} =1.36V		2Ch V _{CORE} =1.68V		3Ch V _{CORE} =2.00V	
0Dh V _{CORE} =1.06V		1Dh V _{CORE} =1.38V		2Dh V _{CORE} =1.70V		3Dh V _{CORE} =2.02V	
0Eh V _{CORE} =1.08V		1Eh V _{CORE} =1.40V		2Eh V _{CORE} =1.72V		3Eh V _{CORE} =2.04V	
0Fh V _{CORE} =1.10V		1Fh V _{CORE} =1.42V		2Fh V _{CORE} =1.74V		3Fh V _{CORE} =2.06V	

VIO (08h) – VIO voltage

7	6	5	4	3	2	1	0
Reserved		V _{IO}					
00h V _{IO} =1.00V		08h V _{IO} =1.40V		10h V _{IO} =1.80V		18h V _{IO} =2.20V	
01h V _{IO} =1.05V		09h V _{IO} =1.45V		11h V _{IO} =1.85V		19h V _{IO} =2.25V	
02h V _{IO} =1.10V		0Ah V _{IO} =1.50V		12h V _{IO} =1.90V		1Ah V _{IO} =2.30V	
03h V _{IO} =1.15V		0Bh V _{IO} =1.55V		13h V _{IO} =1.95V		1Bh V _{IO} =2.35V	
04h V _{IO} =1.20V		0Ch V _{IO} =1.60V		14h V _{IO} =2.00V		1Ch V _{IO} =2.40V	
05h V _{IO} =1.25V		0Dh V _{IO} =1.65V		15h V _{IO} =2.05V		1Dh V _{IO} =2.45V	
06h V _{IO} =1.30V		0Eh V _{IO} =1.70V		16h V _{IO} =2.10V		1Eh V _{IO} =2.50V	
07h V _{IO} =1.35V		0Fh V _{IO} =1.75V		17h V _{IO} =2.15V		1Fh V _{IO} =2.55V	

VGL1 (09h) – VGL1 voltage

7	6	5	4	3	2	1	0
Reserved		V _{GL1}					
00h V _{GL1} =-5.4V		10h V _{GL1} =-8.6V		20h V _{GL1} =-11.8V		30h V _{GL1} =-15.0V	
01h V _{GL1} =-5.6V		11h V _{GL1} =-8.8V		21h V _{GL1} =-12.0V		31h V _{GL1} =-15.2V	
02h V _{GL1} =-5.8V		12h V _{GL1} =-9.0V		22h V _{GL1} =-12.2V		32h V _{GL1} =-15.4V	
03h V _{GL1} =-6.0V		13h V _{GL1} =-9.2V		23h V _{GL1} =-12.4V		33h V _{GL1} =-15.6V	
04h V _{GL1} =-6.2V		14h V _{GL1} =-9.4V		24h V _{GL1} =-12.6V		34h V _{GL1} =-15.8V	
05h V _{GL1} =-6.4V		15h V _{GL1} =-9.6V		25h V _{GL1} =-12.8V		35h V _{GL1} =-16.0V	
06h V _{GL1} =-6.6V		16h V _{GL1} =-9.8V		26h V _{GL1} =-13.0V		36h V _{GL1} =-16.2V	
07h V _{GL1} =-6.8V		17h V _{GL1} =-10.0V		27h V _{GL1} =-13.2V		37h V _{GL1} =-16.4V	
08h V _{GL1} =-7.0V		18h V _{GL1} =-10.2V		28h V _{GL1} =-13.4V		38h V _{GL1} =-16.6V	
09h V _{GL1} =-7.2V		19h V _{GL1} =-10.4V		29h V _{GL1} =-13.6V		39h V _{GL1} =-16.8V	
0Ah V _{GL1} =-7.4V		1Ah V _{GL1} =-10.6V		2Ah V _{GL1} =-13.8V		3Ah V _{GL1} =-17.0V	
0Bh V _{GL1} =-7.6V		1Bh V _{GL1} =-10.8V		2Bh V _{GL1} =-14.0V		3Bh V _{GL1} =-17.2V	
0Ch V _{GL1} =-7.8V		1Ch V _{GL1} =-11.0V		2Ch V _{GL1} =-14.2V		3Ch V _{GL1} =-17.4V	
0Dh V _{GL1} =-8.0V		1Dh V _{GL1} =-11.2V		2Dh V _{GL1} =-14.4V		3Dh V _{GL1} =-17.6V	
0Eh V _{GL1} =-8.2V		1Eh V _{GL1} =-11.4V		2Eh V _{GL1} =-14.6V		3Eh V _{GL1} =-17.8V	
0Fh V _{GL1} =-8.4V		1Fh V _{GL1} =-11.6V		2Fh V _{GL1} =-14.8V		3Fh V _{GL1} =-18.0V	

VCOM (0Ah) – VCOM voltage

7	6	5	4	3	2	1	0
V _{COM}							
00h V _{COM} =-0.00V		01h V _{COM} =0.01V		02h V _{COM} =0.02V		03h V _{COM} =0.03V	
.....							
7Eh V _{COM} =1.26V		7Fh V _{COM} =1.27V		80h V _{COM} =1.28V		81h V _{COM} =1.29V	
.....							
FCh V _{COM} =2.52V		FDh V _{COM} =2.53V		FEh V _{COM} =2.54V		FFh V _{COM} =2.55V	

VCOM_LSB (0Bh) – VCOM_LSB voltage

7	6	5	4	3	2	1	0
Reserved				V _{COM_LSB}			
00h V _{COM_LSB} =0mV		02h V _{COM_LSB} =2.5mV		04h V _{COM_LSB} =5mV		06h V _{COM_LSB} =7.5mV	
01h V _{COM_LSB} =1.25mV		03h V _{COM_LSB} =3.75mV		05h V _{COM_LSB} =6.25mV		07h V _{COM_LSB} =8.75mV	

VRESET (0Ch) – VRESET voltage

V_{RESET} (0Ch) – RESET voltage

7	6	5	4	3	2	1	0
Reserved	V _{DET2}			Reserved	V _{DET1}		
00h V _{RESET2} =2.0V		04h V _{RESET2} =2.4V		00h V _{RESET1} =2.0V		04h V _{RESET1} =2.4V	
01h V _{RESET2} =2.1V		05h V _{RESET2} =2.5V		01h V _{RESET1} =2.1V		05h V _{RESET1} =2.5V	
02h V _{RESET2} =2.2V		06h V _{RESET2} =2.6V		02h V _{RESET1} =2.2V		06h V _{RESET1} =2.6V	
03h V _{RESET2} =2.3V		07h V _{RESET2} =2.7V		03h V _{RESET1} =2.3V		07h V _{RESET1} =2.7V	

VGMA1 (0Dh) – GMA1 voltage

7	6	5	4	3	2	1	0
Reserved		V _{GMA1}					
00h	V _{GMA1} =PVDD-0.1V	10h	V _{GMA1} =PVDD-0.32V	20h	V _{GMA1} =PVDD-0.64V	30h	V _{GMA1} =PVDD-0.96V
01h	V _{GMA1} =PVDD-0.1V	11h	V _{GMA1} =PVDD-0.34V	21h	V _{GMA1} =PVDD-0.66V	31h	V _{GMA1} =PVDD-0.98V
02h	V _{GMA1} =PVDD-0.1V	12h	V _{GMA1} =PVDD-0.36V	22h	V _{GMA1} =PVDD-0.68V	32h	V _{GMA1} =PVDD-1.00V
03h	V _{GMA1} =PVDD-0.1V	13h	V _{GMA1} =PVDD-0.38V	23h	V _{GMA1} =PVDD-0.70V	33h	V _{GMA1} =PVDD-1.02V
04h	V _{GMA1} =PVDD-0.1V	14h	V _{GMA1} =PVDD-0.40V	24h	V _{GMA1} =PVDD-0.72V	34h	V _{GMA1} =PVDD-1.04V
05h	V _{GMA1} =PVDD-0.1V	15h	V _{GMA1} =PVDD-0.42V	25h	V _{GMA1} =PVDD-0.74V	35h	V _{GMA1} =PVDD-1.06V
06h	V _{GMA1} =PVDD-0.12V	16h	V _{GMA1} =PVDD-0.44V	26h	V _{GMA1} =PVDD-0.76V	36h	V _{GMA1} =PVDD-1.08V
07h	V _{GMA1} =PVDD-0.14V	17h	V _{GMA1} =PVDD-0.46V	27h	V _{GMA1} =PVDD-0.78V	37h	V _{GMA1} =PVDD-1.10V
08h	V _{GMA1} =PVDD-0.16V	18h	V _{GMA1} =PVDD-0.48V	28h	V _{GMA1} =PVDD-0.80V	38h	V _{GMA1} =PVDD-1.12V
09h	V _{GMA1} =PVDD-0.18V	19h	V _{GMA1} =PVDD-0.50V	29h	V _{GMA1} =PVDD-0.82V	39h	V _{GMA1} =PVDD-1.14V
0Ah	V _{GMA1} =PVDD-0.20V	1Ah	V _{GMA1} =PVDD-0.52V	2Ah	V _{GMA1} =PVDD-0.84V	3Ah	V _{GMA1} =PVDD-1.16V
0Bh	V _{GMA1} =PVDD-0.22V	1Bh	V _{GMA1} =PVDD-0.54V	2Bh	V _{GMA1} =PVDD-0.86V	3Bh	V _{GMA1} =PVDD-1.18V
0Ch	V _{GMA1} =PVDD-0.24V	1Ch	V _{GMA1} =PVDD-0.56V	2Ch	V _{GMA1} =PVDD-0.88V	3Ch	V _{GMA1} =PVDD-1.20V
0Dh	V _{GMA1} =PVDD-0.26V	1Dh	V _{GMA1} =PVDD-0.58V	2Dh	V _{GMA1} =PVDD-0.90V	3Dh	V _{GMA1} =PVDD-1.22V
0Eh	V _{GMA1} =PVDD-0.28V	1Eh	V _{GMA1} =PVDD-0.60V	2Eh	V _{GMA1} =PVDD-0.92V	3Eh	V _{GMA1} =PVDD-1.24V
0Fh	V _{GMA1} =PVDD-0.30V	1Fh	V _{GMA1} =PVDD-0.62V	2Fh	V _{GMA1} =PVDD-0.94V	3Fh	V _{GMA1} =PVDD-1.26V

VGMA2 (0Eh) – GMA2 voltage

7	6	5	4	3	2	1	0
Reserved		V _{GMA2}					
00h	V _{GMA2} =NVDD+0.1V	10h	V _{GMA2} =NVDD+0.32V	20h	V _{GMA2} =NVDD+0.64V	30h	V _{GMA2} =NVDD+0.96V
01h	V _{GMA2} =NVDD+0.1V	11h	V _{GMA2} =NVDD+0.34V	21h	V _{GMA2} =NVDD+0.66V	31h	V _{GMA2} =NVDD+0.98V
02h	V _{GMA2} =NVDD+0.1V	12h	V _{GMA2} =NVDD+0.36V	22h	V _{GMA2} =NVDD+0.68V	32h	V _{GMA2} =NVDD+1.00V
03h	V _{GMA2} =NVDD+0.1V	13h	V _{GMA2} =NVDD+0.38V	23h	V _{GMA2} =NVDD+0.70V	33h	V _{GMA2} =NVDD+1.02V
04h	V _{GMA2} =NVDD+0.1V	14h	V _{GMA2} =NVDD+0.40V	24h	V _{GMA2} =NVDD+0.72V	34h	V _{GMA2} =NVDD+1.04V
05h	V _{GMA2} =NVDD+0.1V	15h	V _{GMA2} =NVDD+0.42V	25h	V _{GMA2} =NVDD+0.74V	35h	V _{GMA2} =NVDD+1.06V
06h	V _{GMA2} =NVDD+0.12V	16h	V _{GMA2} =NVDD+0.44V	26h	V _{GMA2} =NVDD+0.76V	36h	V _{GMA2} =NVDD+1.08V
07h	V _{GMA2} =NVDD+0.14V	17h	V _{GMA2} =NVDD+0.46V	27h	V _{GMA2} =NVDD+0.78V	37h	V _{GMA2} =NVDD+1.10V
08h	V _{GMA2} =NVDD+0.16V	18h	V _{GMA2} =NVDD+0.48V	28h	V _{GMA2} =NVDD+0.80V	38h	V _{GMA2} =NVDD+1.12V
09h	V _{GMA2} =NVDD+0.18V	19h	V _{GMA2} =NVDD+0.50V	29h	V _{GMA2} =NVDD+0.82V	39h	V _{GMA2} =NVDD+1.14V
0Ah	V _{GMA2} =NVDD+0.20V	1Ah	V _{GMA2} =NVDD+0.52V	2Ah	V _{GMA2} =NVDD+0.84V	3Ah	V _{GMA2} =NVDD+1.16V
0Bh	V _{GMA2} =NVDD+0.22V	1Bh	V _{GMA2} =NVDD+0.54V	2Bh	V _{GMA2} =NVDD+0.86V	3Bh	V _{GMA2} =NVDD+1.18V
0Ch	V _{GMA2} =NVDD+0.24V	1Ch	V _{GMA2} =NVDD+0.56V	2Ch	V _{GMA2} =NVDD+0.88V	3Ch	V _{GMA2} =NVDD+1.20V
0Dh	V _{GMA2} =NVDD+0.26V	1Dh	V _{GMA2} =NVDD+0.58V	2Dh	V _{GMA2} =NVDD+0.90V	3Dh	V _{GMA2} =NVDD+1.22V
0Eh	V _{GMA2} =NVDD+0.28V	1Eh	V _{GMA2} =NVDD+0.60V	2Eh	V _{GMA2} =NVDD+0.92V	3Eh	V _{GMA2} =NVDD+1.24V
0Fh	V _{GMA2} =NVDD+0.30V	1Fh	V _{GMA2} =NVDD+0.62V	2Fh	V _{GMA2} =NVDD+0.94V	3Fh	V _{GMA2} =NVDD+1.26V

PVDD CNTR1 (0Fh)

7	6	5	4	3	2	1	0
LXP_OCP		LXP_SR		LXP_FREQ			
00h	LXP_OCP=1.5A	00h	Fast	00h	LXP_FREQ=600kHz		
01h	LXP_OCP=2.0A	01h	LV2	01h	<u>LXP_FREQ=714kHz</u>		
02h	<u>LXP_OCP=2.5A</u>	02h	LV3	02h	LXP_FREQ=790kHz		
03h	LXP_OCP=3.0A	03h	<u>LV4</u>	03h	LXP_FREQ=937kHz		
		04h	LV5	04h	LXP_FREQ=1000kHz		
		05h	LV6	05h	LXP_FREQ=1250kHz		
		06h	LV7	06h	LXP_FREQ=1250kHz		
		07h	Slow	07h	LXP_FREQ=1250kHz		

PVDD CNTR2 (10h)

7	6	5	4	3	2	1	0
Reserved	PVDD_SS			PVDD_DLY			
	00h	PVDD_SS=2ms		00h	PVDD_DLY=0ms		08h PVDD_DLY=40ms
	01h	PVDD_SS=4ms		01h	PVDD_DLY=5ms		09h PVDD_DLY=45ms
	02h	PVDD_SS=6ms		02h	PVDD_DLY=10ms		0Ah PVDD_DLY=45ms
	03h	PVDD_SS=8ms		03h	PVDD_DLY=15ms		0Bh PVDD_DLY=45ms
	04h	PVDD_SS=10ms		04h	PVDD_DLY=20ms		0Ch PVDD_DLY=45ms
	05h	PVDD_SS=12ms		05h	PVDD_DLY=25ms		0Dh PVDD_DLY=45ms
	06h	PVDD_SS=14ms		06h	PVDD_DLY=30ms		0Eh PVDD_DLY=45ms
	07h	PVDD_SS=16ms		07h	PVDD_DLY=35ms		0Fh PVDD_DLY=45ms

NVDD CNTR1 (11h)

7	6	5	4	3	2	1	0
Reserved	NVDD_SS			NVDD_DLY			
	00h	NVDD_SS=2ms		00h	NVDD_DLY=0ms		08h NVDD_DLY=40ms
	01h	NVDD_SS=4ms		01h	NVDD_DLY=5ms		09h NVDD_DLY=45ms
	02h	NVDD_SS=6ms		02h	NVDD_DLY=10ms		0Ah NVDD_DLY=45ms
	03h	NVDD_SS=8ms		03h	NVDD_DLY=15ms		0Bh NVDD_DLY=45ms
	04h	NVDD_SS=10ms		04h	NVDD_DLY=20ms		0Ch NVDD_DLY=45ms
	05h	NVDD_SS=12ms		05h	NVDD_DLY=25ms		0Dh NVDD_DLY=45ms
	06h	NVDD_SS=14ms		06h	NVDD_DLY=30ms		0Eh NVDD_DLY=45ms
	07h	NVDD_SS=16ms		07h	NVDD_DLY=35ms		0Fh NVDD_DLY=45ms

VGX CNTR1 (12h)

7	6	5	4	3	2	1	0
Reserved			LXH/N_SR		LXH/L_FREQ		
			00h	Fastest	00h	LXH/L_FREQ=500kHz	
			01h	Fast	01h	LXH/L_FREQ=600kHz	
			02h	Normal	02h	LXH/L_FREQ=715kHz	
			03h	Slow	03h	LXH/L_FREQ=800kHz	
					04h	LXH/L_FREQ=933kHz	
					05h	LXH/L_FREQ=1000kHz	
					06h	LXH/L_FREQ=1225kHz	
					07h	LXH/L_FREQ=1430kHz	

VGX CNTR2 (13h)

7	6	5	4	3	2	1	0
SIBO_LFREQ	Reserved	VGH_SS		VGH_DLY			
0	Default	00h	VGH_SS=4ms	00h	VGH_DLY=0ms		08h VGH_DLY=40ms
1	LXHN_Freq half	01h	VGH_SS=8ms	01h	VGH_DLY=5ms		09h VGH_DLY=45ms
		02h	VGH_SS=12ms	02h	VGH_DLY=10ms		0Ah VGH_DLY=45ms
		03h	VGH_SS=16ms	03h	VGH_DLY=15ms		0Bh VGH_DLY=45ms
				04h	VGH_DLY=20ms		0Ch VGH_DLY=45ms
				05h	VGH_DLY=25ms		0Dh VGH_DLY=45ms
				06h	VGH_DLY=30ms		0Eh VGH_DLY=45ms
				07h	VGH_DLY=35ms		0Fh VGH_DLY=45ms

VGX CNTR3 (14h)

7	6	5	4	3	2	1	0
Reserved	VGL_SS			VGL_DLY			
	00h	<u>VGL_SS=4ms</u>		00h	VGL_DLY=0ms		08h VGL_DLY=40ms
	01h	VGL_SS=4ms		01h	VGL_DLY=5ms		09h VGL_DLY=45ms
	02h	VGL_SS=6ms		02h	VGL_DLY=10ms		0Ah VGL_DLY=45ms
	03h	VGL_SS=8ms		03h	<u>VGL_DLY=15ms</u>		0Bh VGL_DLY=45ms
	04h	VGL_SS=10ms		04h	VGL_DLY=20ms		0Ch VGL_DLY=45ms
	05h	VGL_SS=12ms		05h	VGL_DLY=25ms		0Dh VGL_DLY=45ms
	06h	VGL_SS=14ms		06h	VGL_DLY=30ms		0Eh VGL_DLY=45ms
	07h	VGL_SS=16ms		07h	VGL_DLY=35ms		0Fh VGL_DLY=45ms

VCORE CNTR1 (15h)

7	6	5	4	3	2	1	0
Reserved			VCORE_SR		VCORE_FREQ		
			00h	Fastest	00h	VCORE_FREQ=600kHz	
			01h	Fast	01h	<u>VCORE_FREQ=715kHz</u>	
			02h	<u>Normal</u>	02h	VCORE_FREQ=800kHz	
			03h	Slow	03h	VCORE_FREQ=933kHz	
					04h	VCORE_FREQ=1000kHz	
					05h	VCORE_FREQ=1225kHz	
					06h	VCORE_FREQ=1225kHz	
					07h	VCORE_FREQ=1225kHz	

VCORE CNTR2 (16h)

7	6	5	4	3	2	1	0
Reserved	VCORE_SS			VCORE_DLY			
	00h	<u>VCORE_SS=2ms</u>		00h	<u>VCORE_DLY=0ms</u>		08h VCORE_DLY=40ms
	01h	VCORE_SS=4ms		01h	VCORE_DLY=5ms		09h VCORE_DLY=45ms
	02h	VCORE_SS=6ms		02h	VCORE_DLY=10ms		0Ah VCORE_DLY=45ms
	03h	VCORE_SS=8ms		03h	VCORE_DLY=15ms		0Bh VCORE_DLY=45ms
				04h	VCORE_DLY=20ms		0Ch VCORE_DLY=45ms
				05h	VCORE_DLY=25ms		0Dh VCORE_DLY=45ms
				06h	VCORE_DLY=30ms		0Eh VCORE_DLY=45ms
				07h	VCORE_DLY=35ms		0Fh VCORE_DLY=45ms

VIO CNTR1 (17h)

7	6	5	4	3	2	1	0
Reserved	VIO_SR			VIO_FREQ			
			00h	Fastest	00h	VIO_FREQ=600kHz	
			01h	Fast	01h	<u>VIO_FREQ=715kHz</u>	
			02h	<u>Normal</u>	02h	VIO_FREQ=800kHz	
			03h	Slow	03h	VIO_FREQ=933kHz	
					04h	VIO_FREQ=1000kHz	
					05h	VIO_FREQ=1225kHz	
					06h	VIO_FREQ=1225kHz	
					07h	VIO_FREQ=1225kHz	

VIO_CNTR2 (18h)

7	6	5	4	3	2	1	0
Reserved		VIO_SS		VIO_DLY			
		00h	VCORE_SS=2ms	00h	VIO_DLY=0ms	08h	VIO_DLY=40ms
		01h	VCORE_SS=4ms	01h	VIO_DLY=5ms	09h	VIO_DLY=45ms
		02h	VCORE_SS=6ms	02h	VIO_DLY=10ms	0Ah	VIO_DLY=45ms
		03h	VCORE_SS=8ms	03h	VIO_DLY=15ms	0Bh	VIO_DLY=45ms
				04h	VIO_DLY=20ms	0Ch	VIO_DLY=45ms
				05h	VIO_DLY=25ms	0Dh	VIO_DLY=45ms
				06h	VIO_DLY=30ms	0Eh	VIO_DLY=45ms
				07h	VIO_DLY=35ms	0Fh	VIO_DLY=45ms

RESET_DLY (19h)

7	6	5	4	3	2	1	0
RESET2_DLY				RESET1_DLY			
00h	RESET2_DLY=0ms	08h	RESET2_DLY=40ms	00h	RESET1_DLY=0ms	08h	RESET1_DLY=40ms
01h	RESET2_DLY=5ms	09h	RESET2_DLY=45ms	01h	RESET1_DLY=5ms	09h	RESET1_DLY=45ms
02h	RESET2_DLY=10ms	0Ah	RESET2_DLY=45ms	02h	RESET1_DLY=10ms	0Ah	RESET1_DLY=45ms
03h	RESET2_DLY=15ms	0Bh	RESET2_DLY=45ms	03h	RESET1_DLY=15ms	0Bh	RESET1_DLY=45ms
04h	RESET2_DLY=20ms	0Ch	RESET2_DLY=45ms	04h	RESET1_DLY=20ms	0Ch	RESET1_DLY=45ms
05h	RESET2_DLY=25ms	0Dh	RESET2_DLY=45ms	05h	RESET1_DLY=25ms	0Dh	RESET1_DLY=45ms
06h	RESET2_DLY=30ms	0Eh	RESET2_DLY=45ms	06h	RESET1_DLY=30ms	0Eh	RESET1_DLY=45ms
07h	RESET2_DLY=35ms	0Fh	RESET2_DLY=45ms	07h	RESET1_DLY=35ms	0Fh	RESET1_DLY=45ms

VCOM_CNTR (1Bh)

7	6	5	4	3	2	1	0
Reserved		VCOM_OFF_SEL	VCOM_DLY				
		00h	UVLO_F	00h	VCOM_DLY=0ms	08h	VCOM_DLY=55ms
		01h	VDET	01h	VCOM_DLY=5ms	0Ch	VCOM_DLY=60ms
				02h	VCOM_DLY=10ms	0Dh	VCOM_DLY=65ms
				03h	VCOM_DLY=15ms	0Eh	VCOM_DLY=70ms
				04h	VCOM_DLY=20ms	0Fh	VCOM_DLY=75ms
				05h	VCOM_DLY=25ms	10h	VCOM_DLY=80ms
				06h	VCOM_DLY=30ms	11h	VCOM_DLY=85ms
				07h	VCOM_DLY=35ms	12h	VCOM_DLY=90ms
				08h	VCOM_DLY=40ms	13h	VCOM_DLY=95ms
				09h	VCOM_DLY=45ms	14h	VCOM_DLY=100ms
				0Ah	VCOM_DLY=50ms	15h	VCOM_DLY=105ms
						16h	VCOM_DLY=110ms
						17h	VCOM_DLY=115ms
						18h	VCOM_DLY=120ms
						19h	VCOM_DLY=125ms
						1Ah	VCOM_DLY=130ms
						1Bh	VCOM_DLY=135ms
						1Ch	VCOM_DLY=140ms
						1Dh	VCOM_DLY=145ms
						1Eh	VCOM_DLY=150ms
						1Fh	VCOM_DLY=155ms

NVDD_CNTR2 (1Ch)

7	6	5	4	3	2	1	0
Reserved		LXN_SR		LXN_OCP		Reserved	
		00h	Fastest	00h	LXN_OCP=2.5A		
		01h	Fast	01h	LXN_OCP=2.0A		
		02h	Normal	02h	LXN_OCP=1.5A		
		03h	Slow	03h	LXN_OCP=1.0A		

VCOM_MIN (1Dh)

7	6	5	4	3	2	1	0
Reserved			VCOM_MIN				
00h	VCOM_MIN=-3.60V	08h	VCOM_MIN=-2.70V	10h	VCOM_MIN=-1.50V	18h	VCOM_MIN=-0.30V
01h	VCOM_MIN=-3.60V	09h	VCOM_MIN=-2.55V	11h	VCOM_MIN=-1.35V	19h	VCOM_MIN=-0.15V
02h	VCOM_MIN=-3.60V	0Ah	VCOM_MIN=-2.40V	12h	VCOM_MIN=-1.20V	1Ah	VCOM_MIN=-0.00V
03h	VCOM_MIN=-3.45V	0Bh	VCOM_MIN=-2.25V	13h	VCOM_MIN=-1.05V	1Bh	VCOM_MIN=0.15V
04h	VCOM_MIN=-3.30V	0Ch	VCOM_MIN=-2.10V	14h	VCOM_MIN=-0.90V	1Ch	VCOM_MIN=0.30V
05h	VCOM_MIN=-3.15V	0Dh	VCOM_MIN=-1.95V	15h	VCOM_MIN=-0.75V	1Dh	VCOM_MIN=0.45V
06h	VCOM_MIN=-3.00V	0Eh	VCOM_MIN=-1.80V	16h	VCOM_MIN=-0.60V	1Eh	VCOM_MIN=0.60V
07h	VCOM_MIN=-2.85V	0Fh	<u>VCOM_MIN=-1.65V</u>	17h	VCOM_MIN=-0.45V	1Fh	VCOM_MIN=0.75V

RCOMP (1Fh)

7	6	5	4	3	2	1	0
VGL_RCOMP		VGH_RCOMP		NVDD_RCOMP		PVDD_RCOMP	
00h	VGL_RCOMP=133%	00h	VGH_RCOMP=56%	00h	NVDD_RCOMP=167%	00h	PVDD_RCOMP=33%
01h	VGL_RCOMP=116%	01h	VGH_RCOMP=37%	01h	NVDD_RCOMP=133%	01h	PVDD_RCOMP=67%
02h	<u>VGL_RCOMP=100%</u>	02h	<u>VGH_RCOMP=100%</u>	02h	<u>NVDD_RCOMP=100%</u>	02h	<u>PVDD_RCOMP=100%</u>
03h	VGL_RCOMP=83%	03h	VGH_RCOMP=75%	03h	NVDD_RCOMP=67%	03h	PVDD_RCOMP=133%

RCOMP (20h)

7	6	5	4	3	2	1	0
Reserved				VCORE_RCOMP		VIO_RCOMP	
				00h	VCORE_RCOMP=133%	00h	VIO_RCOMP=133%
				01h	<u>VCORE_RCOMP=100%</u>	01h	<u>VIO_RCOMP=100%</u>
				02h	VCORE_RCOMP=67%	02h	VIO_RCOMP=67%
				03h	VCORE_RCOMP=33%	03h	VIO_RCOMP=33%

Option1 (21h)

7	6	5	4	3	2	1	0
BUCK_BUF_RD		VGH_LT_SEL		OTP_EN	UVP_EN	ABN_EN	
00h	Always	00h	VGH_LT_SEL=2V	00h	Disable	00h	Disable
01h	After I2C CMD	01h	VGH_LT_SEL=4V	01h	<u>Enable</u>	01h	infinite Recovery
10h	<u>After Memory_W</u>	10h	<u>VGH_LT_SEL=6V</u>			10h	3 times Recovery
11h	Only Power on reset	11h	VGH_LT_SEL=8V			11h	<u>Latch</u>

Option2 (22h)

7		6		5		4		3		2		1		0	
UBRR_IN		SIBO_EN		VG_PHASE				VGH_preSS_5ms		VGH_PRECHG		VGL2_OCPH		VCOM_BIAS	
00h	<u>READ_OK & ALL IP SSF</u>	00h	<u>Over 50%</u>	00h	default -2*step				00h	<u>1ms</u>	00h	<u>Disable</u>	00h	<u>OCP=0.75A</u>	
		01h	<u>Under 50%</u>	01h	default -1*step				01h	5ms	01h	Enable	01h	OCP=1.5A	
01h	READ_OK			02h	<u>default (50%)</u>										
				03h	default +1*step										

Note

- 1) SIBO_EN: VGL2 Max Duty
- 2) VG_PHASE: VGH set to VGL2 set Phase Control
- 3) VGH_preSS_5ms: (VGH_SLPb_LH rise) to (VGH_SSCNT start)

UBRR CTRL1 (23h)

7	6	5	4	3	2	1	0
Reserved				EN_VGL1_UBRR3	EN_VGL2_UBRR3	EN_NVDD_UBRR3	EN_PVDD_UBRR3
				00h Disable	00h Disable	00h Disable	00h Disable
				01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>

UBRR1 CTRL2 (24h)

7	6	5	4	3	2	1	0
Reserved				UBRR_IN4	UBRR_IN3	UBRR_IN2	UBRR_IN1
				00h <u>Normal</u>	00h <u>Normal</u>	00h <u>Normal</u>	00h <u>Normal</u>
				01h UBRR4	01h UBRR3	01h UBRR2	01h UBRR1

UBRR CTRL3 (25h)

7	6	5	4	3	2	1	0
Reserved				iCOMP_Use		iCTRL_Use	
				00h RNTC OFF		00h CTRL INPUT	
				01h <u>RNTC ON</u>		01h <u>UBRR1 INPUT</u>	
				02h UBRR2 User		02h VDET2 OUTPUT	
				03h None		03h UBRR mode detect change	

UBRR CTRL4 (26h)

7	6	5	4	3	2	1	0
EN_VCOM_UBRR1	EN_VGL1_UBRR1	EN_VCORE_UBRR1	EN_VIO_UBRR1	EN_VGL2_UBRR1	EN_VGH_UBRR1	EN_NVDD_UBRR1	EN_PVDD_UBRR1
00h Disable	00h Disable	00h Disable	00h Disable	00h Disable	00h Disable	00h Disable	00h Disable
01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>

UBRR CTRL5 (27h)

7	6	5	4	3	2	1	0
EN_VCORE_UBRR3	EN_VIO_UBRR3	EN_RESET1_UBRR1	EN_RESET2_UBRR1	EN_GMA_UBRR1	UBRR1_DISCH_EN	PVDD_DOWN1	NVDD_DOWN1
00h <u>Disable</u>	00h <u>Disable</u>	00h Disable	00h Disable	00h Disable	00h Disable	00h Disable	00h Disable
01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>

UBRR CTRL6 (28h)

7	6	5	4	3	2	1	0
PVDD_DLV				NVDD_DLV			
00h <u>PVDD_DLV=4.0V</u>	08h PVDD_DLV=4.8V			00h <u>NVDD_DLV=-4.0V</u>	08h NVDD_DLV=-4.8V		
01h PVDD_DLV=4.1V	09h PVDD_DLV=4.9V			01h NVDD_DLV=-4.1V	09h NVDD_DLV=-4.9V		
02h PVDD_DLV=4.2V	0Ah PVDD_DLV=5.0V			02h NVDD_DLV=-4.2V	0Ah NVDD_DLV=-5.0V		
03h PVDD_DLV=4.3V	0Bh PVDD_DLV=5.1V			03h NVDD_DLV=-4.3V	0Bh NVDD_DLV=-5.1V		
04h PVDD_DLV=4.4V	0Ch PVDD_DLV=5.2V			04h NVDD_DLV=-4.4V	0Ch NVDD_DLV=-5.2V		
05h PVDD_DLV=4.5V	0Dh PVDD_DLV=5.3V			05h NVDD_DLV=-4.5V	0Dh NVDD_DLV=-5.3V		
06h PVDD_DLV=4.6V	0Eh PVDD_DLV=5.4V			06h NVDD_DLV=-4.6V	0Eh NVDD_DLV=-5.4V		
07h PVDD_DLV=4.7V	0Fh PVDD_DLV=5.5V			07h NVDD_DLV=-4.7V	0Fh NVDD_DLV=-5.5V		

UBRR CTRL7 (29h)

7	6	5	4	3	2	1	0
EN_VCOM_UBRR2	EN_VGL1_UBRR2	EN_VCORE_UBRR2	EN_VIO_UBRR2	EN_VGL2_UBRR2	EN_VGH_UBRR2	EN_NVDD_UBRR2	EN_PVDD_UBRR2
00h Disable	00h Disable	00h Disable	00h Disable	00h Disable	00h Disable	00h Disable	00h Disable
01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>

UBRR CTRL8 (2Ah)

7	6	5	4	3	2	1	0
NVDD_DOWN3	PVDD_DOWN3	EN_RESET1_UBRR2	EN_RESET2_UBRR2	EN_GMA_UBRR2	UBRR2_DISCH_EN	PVDD_DOWN2	NVDD_DOWN2
00h Disable	00h Disable	00h Disable	00h Disable	00h Disable	00h Disable	00h Disable	00h Disable
01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>	01h <u>Enable</u>

Control (FEh)

7	6	5	4	3	2	1	0
Reserved				WED_VCOM	Reserved		
WED_VCOM							
Setting this bit forces the contents of VCOM Voltage Registers to be copied into EEPROM, thereby marking them the default values during power up.							
When the contents of VCOM Voltage Registers have been written to the EEPROM, the IC automatically resets this bit.							

Control (FFh)

7	6	5	4	3	2	1	0
WED_ALL	Reserved						RED
WED_ALL	Setting this bit forces the contents of all Registers to be copied into EEPROM, thereby marking them the default values during power up.						
	When the contents of all Registers have been written to the EEPROM, the IC automatically resets this bit.						
RED	0 Read operations return the contents of the Registers						
	1 Read operations return the contents of the EEPROM						

VCOM (00h) – VCOM voltage : Slave 50h

7	6	5	4	3	2	1	0
V _{COM}							
00h V _{COM} =-0.00V	01h V _{COM} =0.01V		02h V _{COM} =0.02V		03h V _{COM} =0.03V		
.....							
7Eh V _{COM} =1.26V	7Fh V _{COM} =1.27V		80h <u>V_{COM}=1.28V</u>		81h V _{COM} =1.29V		
.....							
FC h V _{COM} =2.52V	FDh V _{COM} =2.53V		FEh V _{COM} =2.54V		FFh V _{COM} =2.55V		

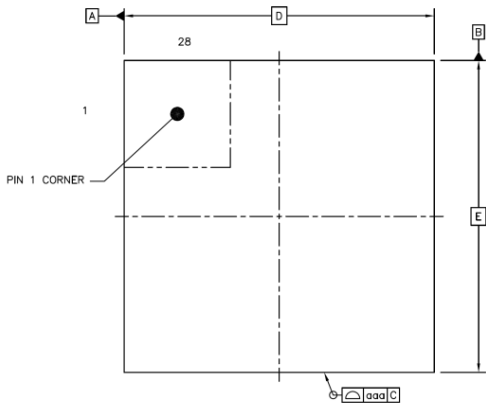
VCOM_LSB (01h) – VCOM_LSB voltage : Slave 50h

7	6	5	4	3	2	1	0
Reserved				V _{COM_LSB}			
00h <u>V_{COM_LSB}=0mV</u>	02h V _{COM_LSB} =2.5mV		04h V _{COM_LSB} =5mV		06h V _{COM_LSB} =7.5mV		
01h V _{COM_LSB} =1.25mV	03h V _{COM_LSB} =3.75mV		05h V _{COM_LSB} =6.25mV		07h V _{COM_LSB} =8.75mV		

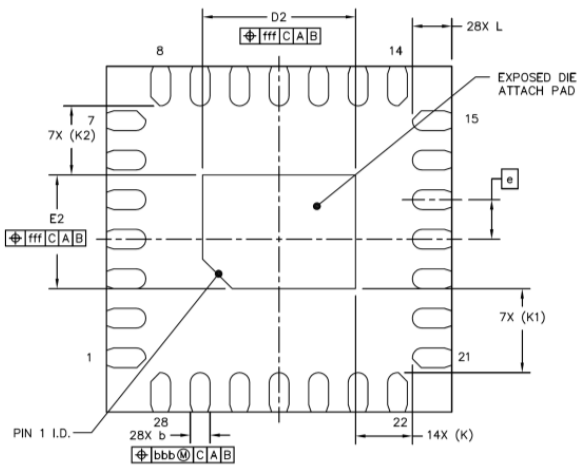
PVCOM_SEL (02h) – PVCOM selection : Slave 50h

7	6	5	4	3	2	1	0
PVCOM SEL	Reserved						
PVCOM SEL	0 : Memory & Register Write / Memory Read						
	1 : Register Write / Register Read						

Package Information



		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.5	0.55	0.6
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	---	0.4	---
L/F THICKNESS		A3	0.152 REF		
LEAD WIDTH		b	0.15	0.2	0.25
BODY SIZE	X	D	3.5 BSC		
	Y	E	3.5 BSC		
LEAD PITCH		e	0.4 BSC		
EP SIZE	X	D2	1.5	1.55	1.6
	Y	E2	1.1	1.15	1.2
LEAD LENGTH		L	0.35	0.4	0.45
LEAD TIP TO EXPOSED PAD EDGE		K	0.575 REF		
		K1	0.85 REF		
		K2	0.7 REF		
PACKAGE EDGE TOLERANCE		aaa	0.1		
MOLD FLATNESS		ccc	0.1		
COPLANARITY		eee	0.08		
LEAD OFFSET		bbb	0.07		
EXPOSED PAD OFFSET		fff	0.1		



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